Highlights

Multi-Task Particle Swarm Optimization with Knowledge-Guided Elite Refinement and Two-Phase Weight Scheduling for High-Dimensional Feature Selection

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- Closed-loop knowledge-guided swarm learning framework for high-dimensional feature selection.
- MI initialization with adaptive re-initialization restores diversity and prevents search entrapment.
- Knowledge-guided elite refinement accumulates feature-importance signals with exponential decay.
- Two-phase weight scheduling temporally decouples feature compression from accuracy refinement.
- KDS-EL achieves 95.74% accuracy with 87.8 selected features across 20 benchmark datasets.

Multi-Task Particle Swarm Optimization with Knowledge-Guided Elite Refinement and Two-Phase Weight Scheduling for High-Dimensional Feature Selection

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ABSTRACT

Feature selection in high-dimensional data remains a central challenge for evolutionary computation. Particle swarm optimization suffers from three structural deficiencies: random initialization provides insufficient guidance in sparse high-dimensional spaces, uniform feature perturbation fails to exploit cross-iteration feature-importance signals, and static objective preferences cannot adapt to different evolutionary phases. To address these limitations, we propose KDS-EL (Knowledge-Guided Dual-Phase Swarm Evolutionary Learning), a knowledge-guided evolutionary framework with a closed-loop feature-importance learning mechanism. First, to overcome inefficient initialization, we propose mutual-information-guided initialization with adaptive re-initialization, injecting prior knowledge and recovering population diversity when stagnation occurs. Second, to exploit discarded cross-iteration feature-importance signals, we propose a knowledge-guided elite refinement strategy that accumulates feature-importance knowledge and steers elite solutions toward historically effective features. Third, to align objective weights with search phases, we propose two-phase weight scheduling that emphasizes early dimensionality reduction and late-stage accuracy refinement. On 20 high-dimensional benchmarks, KDS-EL achieves the best classification performance among all competitors, with a mean accuracy of 0.9574 and only 87.8 selected features, outperforming the strongest single- and multi-objective competitors by +3.97 and +4.42 percentage points; Wilcoxon signed-rank tests with Holm–Bonferroni correction confirm statistically significant improvements over all 13 competitors.

1. Introduction

High-dimensional data have emerged as a fundamental challenge in classification problems across diverse domains, including genomics, text mining, and computer vision. These datasets typically contain hundreds or even tens of thousands of features, while only a limited number of labeled samples are available, resulting in the well-known $p \gg n$ scenario. Under such conditions, conventional learning algorithms often suffer from severe overfitting, increased computational complexity, and reduced interpretability due to the curse of dimensionality [1, 2]. Feature selection (FS) has therefore become an essential preprocessing technique that aims to identify a compact and informative subset of features while preserving classification performance. However, selecting an optimal feature subset remains highly challenging because the search space grows exponentially with the number of features D, and the underlying subset-selection problem is NP-hard [3]. Moreover, individual feature relevance measures are insufficient to characterize complex feature interactions and dependencies, which are often critical for achieving robust classification performance.

FS methods are generally categorized into filter, wrapper, and embedded approaches [1, 4]. Filter methods rank features according to statistical criteria and offer high computational efficiency, but they usually evaluate features independently of the classifier and may overlook feature interactions relevant to classification. Embedded methods incorporate feature selection into the learning process, providing a compromise between selection efficiency and predictive performance; however, their effectiveness is often dependent on the characteristics of specific learning models. Wrapper methods directly optimize feature subsets according to classifier performance and can achieve competitive classification results, but the repeated model evaluations make them computationally expensive, especially in high-dimensional search spaces. To address the combinatorial complexity of wrapper-based FS, evolutionary computation has been extensively investigated as a promising optimization paradigm [3, 5–7]. By maintaining a population of candidate solutions, evolutionary algorithms can perform global exploration of feature subsets without requiring gradient information. Among them, particle swarm optimization (PSO), genetic algorithms, and differential evolution have demonstrated promising results on high-dimensional FS tasks.

PSO has become one of the most widely adopted evolutionary algorithms for FS due to its simple structure, efficient information-sharing mechanism, and fast convergence. Nevertheless, conventional PSO-based FS methods still encounter three structural limitations when dealing with sparse and high-dimensional search spaces. First, random initialization distributes particles without prior guidance throughout the binary search space, whereas effective feature subsets usually occupy only a small region with limited active dimensions. Consequently, a considerable number of fitness evaluations may be wasted on redundant and suboptimal solutions before the swarm identifies promising search regions. Second, conventional uniform feature perturbation strategies treat all dimensions equally and neglect the feature-level improvement information accumulated during previous evaluations, preventing the search from exploiting historically informative features. Third, existing PSO-based FS methods typically employ fixed objective weights to balance classification error and feature subset size during the entire optimization process. However, the desired trade-off between feature reduction and accuracy refinement varies across search stages: aggressive sparsity enhancement is preferred in early exploration, whereas accurate decision boundary refinement becomes more important after a compact subset has been obtained.

Recent evolutionary FS studies have attempted to improve search performance through enhanced population initialization, adaptive evolutionary operators, and refined fitness evaluation strategies [8–11]. Although these approaches have achieved promising results, they usually focus on isolated components of the optimization process and do not jointly address inefficient initialization, insufficient exploitation of accumulated feature knowledge, and inflexible objective weighting. Recently, multi-task PSO has emerged as an effective paradigm for high-dimensional FS with fixed-length binary representations. For instance, MEL [12] adopts cooperative subpopulations and a fixed-length representation to efficiently handle feature spaces containing tens of thousands of dimensions. Despite its effectiveness, existing multi-task PSO-based FS frameworks still leave several challenges unresolved: the initial population is generally generated without sufficient feature-level prior knowledge, useful feature-importance information discovered during evolution is not explicitly retained and exploited, and the balance between classification performance and feature sparsity remains unchanged across different search stages.

To address these limitations, this paper proposes Knowledge-Guided Dual-Phase Swarm Evolutionary Learning (KDS-EL), a closed-loop framework that cohesively integrates domain prior injection, cross-iteration knowledge accumulation, and dynamic objective scheduling into a cooperative multi-task PSO backbone.

The main contributions of this paper are summarized as follows:

1. *MI-Guided Initialization with Adaptive Re-initialization*: A hybrid seeding strategy that injects continuous feature-relevance priors into population construction to eliminate early-stage search waste, coupled with an adaptive re-initialization mechanism that restores swarm diversity upon evolutionary stagnation (Section 3.2).
2. *Knowledge-Guided Elite Refinement (KER)*: A targeted elite perturbation operator that dynamically harvests cross-iteration feature-importance feedback with exponential memory decay, directing elite solutions toward historically effective features within a strictly bounded evaluation overhead (Section 3.4).

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Table 1: Taxonomic and methodological comparison between KDS-EL and representative evolutionary feature selection algorithms.

		Knowledge Engineering			Search Control		
Algorithm	Paradigm	MI Prior Seeding	Online Memory	Memory Decay	Elite Refinement	Dynamic Weighting	Complexity
BPSO [7]	SO-SI	×	×	×	×	×	High
FWPSO [8]	SO-SI	○	×	×	×	×	Moderate
SaWDE [9]	SO-DE	×	×	×	×	○	High
W-MOPSO-FS [13]	MO-SI	×	×	×	×	×	High
NNS-DFPG-CSA [14]	SO-CSA	○	×	×	×	×	Moderate
MI-MPODE [15]	SO-DE	✓	×	×	×	×	Moderate
EMSWOA [16]	MS-WOA	×	×	×	○	×	High
MEL [12]	MT-SI	×	✓	×	×	×	High
KDS-EL (Proposed)	**MT-SI**	✓	✓	✓	✓	✓	**Low**

✓: Fully supported; ○: Partially supported; ×: Absent / not supported.
Paradigms: SO: Single-Objective; MO: Multi-Objective; MS: Multi-Swarm; MT: Multi-Task; SI: Swarm Intelligence; DE: Differential Evolution; CSA: Crow Search Algorithm; WOA: Whale Optimization Algorithm.

3. *Two-Phase Weight Scheduling (TPS)*: A stage-aware objective scheduling mechanism that temporally decouples early aggressive feature pruning from late-stage accuracy refinement, overcoming the rigidity of static weighting (Section 3.5).
4. *Comprehensive Empirical Validation*: Extensive evaluations across 20 high-dimensional benchmark datasets (spanning 500 to 54,675 features) against 13 state-of-the-art competitors, rigorously validated through non-parametric statistical testing, component contribution analysis, and parametric sensitivity investigations (Section 4).

Extensive experiments demonstrate that KDS-EL achieves the best classification performance, obtaining an average classification accuracy of 0.9574 with only 87.8 selected features. Compared with the strongest single-objective and multi-objective competitors, KDS-EL improves accuracy by 3.97 and 4.42 percentage points, respectively, while achieving a 3.4× speedup over standard MEL [12] (reducing average runtime from 175.0 s to 51.4 s). Furthermore, Wilcoxon signed-rank tests with Holm–Bonferroni correction confirm that the improvements are statistically significant over all 13 competitors.

The remainder of this paper is organized as follows. Section 2 formulates the feature selection optimization problem and reviews related literature and background concepts. Section 3 presents the proposed KDS-EL framework and formulates its three constituent mechanisms in detail. Section 4 presents comprehensive experimental results, statistical tests, component contribution analysis, and parameter sensitivity investigations. Finally, Section 5 concludes the paper and discusses prospective research directions.

2. Related Work and Background

This section provides a systematic survey of evolutionary feature selection paradigms, establishes the formal mathematical model of feature selection, formulates continuous and binary particle swarm optimization, reviews mutual information for feature relevance estimation, reviews the Multi-Task Evolutionary Learning (MEL) algorithm, and defines the unique taxonomic positioning of KDS-EL.

2.1. Problem formulation of feature selection

Feature selection (FS) is a combinatorial optimization problem that aims to identify an optimal, minimal subset of informative features from an original D-dimensional feature space, simultaneously maximizing classification performance and eliminating irrelevant or redundant dimensions [1, 2].

Let $\mathcal{D} = \{(\mathbf{f}_i, y_i)\}_{i=1}^{N_{\text{samp}}}$ denote a supervised dataset comprising N_{samp} training instances, where $\mathbf{f}_i = [f_{i,1}, f_{i,2}, \ldots, f_{i,D}] \in \mathbb{R}^D$ is the full feature vector and $y_i \in \{1, 2, \ldots, C\}$ is the corresponding discrete class label across C categories. A candidate feature subset is mathematically represented as a binary decision vector $\mathbf{x} = [x_1, x_2, \ldots, x_D] \in \{0, 1\}^D$, defined as:

$$x_j = \begin{cases} 1, & \text{if feature } f_j \text{ is selected,} \\ 0, & \text{if feature } f_j \text{ is discarded.} \end{cases} \tag{1}$$

The feature selection task inherently balances two dual criteria: minimizing classification error rate while minimizing the feature selection ratio (FSR). In evolutionary wrapper feature selection, this is mathematically formulated as minimizing a unified scalar objective function $f(\mathbf{x})$ [14, 16]:

$$\min_{\mathbf{x}\in\{0,1\}^D} f(\mathbf{x}) = w_a \cdot (1 - \text{Acc}(\mathbf{x})) + w_f \cdot \left(\frac{\|\mathbf{x}\|_0}{D}\right), \tag{2}$$

where $\text{Acc}(\mathbf{x})$ denotes the cross-validation classification accuracy evaluated on the masked feature subset $\{\mathbf{f}_i \odot \mathbf{x}\}_{i=1}^{N_{\text{samp}}}$ using an inductive learning algorithm, $\|\mathbf{x}\|_0 = \sum_{j=1}^{D} x_j$ denotes the L_0-pseudo-norm representing the number of active features, and $w_a \in [0, 1], w_f = 1 - w_a$ are weighting parameters balancing classification accuracy and subset compactness.

Because the cardinality of the discrete search space is $|\Omega| = 2^D$, exhaustive evaluation is computationally intractable for high-dimensional biomedical regimes ($D \gg 10^3$). Traditional feature selection techniques are categorized into three classical paradigms: filter, wrapper, and embedded methods [2]. *Filter methods* evaluate features based on intrinsic statistical properties, such as mutual information (mRMR [17]), correlation (FCBF [18]), or centrality measures [19]. *Embedded methods* perform feature selection during model training, such as Bolasso [20] and tree-based ensemble importance (FS-GBDT [21]). *Wrapper methods* evaluate candidate subsets using the direct predictive feedback of a classifier [4]. While wrappers capture intricate feature interactions and deliver superior accuracy, navigating the 2^D combinatorial space requires efficient stochastic global search heuristics, for which Evolutionary Computation (EC) paradigms—including Genetic Algorithms (GA) [22], Ant Colony Optimization (ACO) [23], Artificial Bee Colony (ABC) [24], and Swarm Intelligence (SI) [3, 25]—have emerged as dominant approaches.

To combat the curse of dimensionality in small-sample high-dimensional regimes, numerous specialized single-objective evolutionary algorithms have been developed: self-adaptive swarm mechanisms [26], progressively refined search trajectories [27], self-adaptive differential evolution with adaptive weighting (SaWDE) [9], multi-layer perceptron-guided PSO (IGPSO) [10], multi-criterion scoring PSO (PSO-CSM) [11], reinforcement learning-based swarm control [28], and divide-and-conquer subspace decomposition [29]. While these methods enhance local search operators, they universally treat feature perturbation as a memoryless stochastic trial, discarding the accumulated evolutionary feedback across generations.

2.2. Particle swarm optimization

PSO, originally introduced by Kennedy and Eberhart [5], is a population-based stochastic optimization paradigm inspired by the collective foraging behaviors of bird flocks. A swarm consists of N particles navigating a D-dimensional continuous search space $\mathbb{R}^D$. Each particle $i \in \{1, \ldots, N\}$ maintains a continuous position vector $\mathbf{p}_i^{(t)} = [p_{i,1}^{(t)}, \ldots, p_{i,D}^{(t)}]$ and a continuous velocity vector $\mathbf{v}_i^{(t)} = [v_{i,1}^{(t)}, \ldots, v_{i,D}^{(t)}]$ at iteration t. The historical trajectory is governed by its personal best position $\mathbf{p}_{\text{best},i}$ and the swarm's global best position $\mathbf{g}_{\text{best}}$.

The velocity and position updates are formulated as:

$$\mathbf{v}_i^{(t+1)} = \omega \mathbf{v}_i^{(t)} + c_1 r_1 (\mathbf{p}_{\text{best},i} - \mathbf{p}_i^{(t)}) + c_2 r_2 (\mathbf{g}_{\text{best}} - \mathbf{p}_i^{(t)}), \tag{3}$$

$$\mathbf{p}_i^{(t+1)} = \mathbf{p}_i^{(t)} + \mathbf{v}_i^{(t+1)}, \tag{4}$$

where ω denotes the inertia weight governing momentum retention [6], c_1 and c_2 are cognitive and social acceleration coefficients, and $r_1, r_2 \sim \mathcal{U}(0, 1)$ are independent uniform random stochastic factors.

To adapt PSO to combinatorial binary feature selection where candidate solutions represent binary indicator masks $\mathbf{x}_i \in \{0,1\}^D$, Kennedy and Eberhart [7] proposed the standard Binary PSO (BPSO). BPSO transforms continuous velocities into selection probabilities via a Sigmoid transfer function:

$$S(v_{i,j}^{(t+1)}) = \frac{1}{1+\exp(-v_{i,j}^{(t+1)})}, \quad x_{i,j}^{(t+1)} = \begin{cases} 1, & \text{if rand}() < S(v_{i,j}^{(t+1)}), \\ 0, & \text{otherwise.} \end{cases} \tag{5}$$

While effective for low-dimensional problems, standard BPSO suffers severe exploration-exploitation dilemmas in ultra-high-dimensional spaces: continuous velocity updates provide no domain-grounded guidance toward sparse, informative biomarker subspaces. In KDS-EL, particles maintain continuous normalized position coordinates $\mathbf{p}_i \in [0,1]^D$, mapped deterministically to binary feature masks via a strict thresholded Heaviside step function $x_{i,j} = \mathbb{I}(p_{i,j} \geq \theta)$ with $\theta = 0.60$, eliminating stochastic binarization variance.

2.3. Mutual information

Mutual information (MI), rooted in Shannon's information theory [30], quantifies the mutual dependence and reduction in uncertainty between two random variables. For a feature variable f_j and the discrete class label $Y \in \{1,\ldots,C\}$, the mutual information $I(f_j;Y)$ is defined as:

$$\begin{aligned} I(f_j;Y) &= H(Y) - H(Y \mid f_j) \\ &= \sum_{v \in \text{Val}(f_j)} \sum_{c=1}^{C} P(f_j = v, Y = c) \log_2 \frac{P(f_j = v, Y = c)}{P(f_j = v)P(Y = c)}, \end{aligned} \tag{6}$$

where $H(Y)$ is the marginal Shannon entropy of the class labels and $H(Y \mid f_j)$ is the conditional entropy given feature f_j. High values of $I(f_j;Y)$ indicate strong discriminative relevance to classification targets.

Conventional filter feature selection methods [14, 17] evaluate $I(f_j;Y)$ across all D dimensions and execute a hard truncation:

$$\mathcal{S}_{\text{Filter}} = \{\text{IDX}(1), \text{IDX}(2), \ldots, \text{IDX}(\lfloor k \cdot D \rfloor)\}, \tag{7}$$

where IDX denotes the feature indices sorted in descending order of $I(f_j;Y)$ and $k \in (0,1]$ represents the retention ratio, permanently discarding all features outside the top-k ranking. While computationally rapid ($O(D \log D)$), hard truncation fundamentally ignores multi-feature interactions: individual features with moderate MI scores that are collectively decisive when combined are permanently eliminated. Rather than using hard truncation, KDS-EL utilizes normalized MI scores as a soft prior to construct continuous initial population seeds (Section 3.2), injecting filter knowledge into the evolutionary wrapper without irreversibly shrinking the search space.

2.4. Multi-task evolutionary learning

Evolutionary Multi-Task Optimization (EMTO) aims to solve multiple distinct optimization tasks simultaneously by facilitating inter-task knowledge transfer through a unified chromosomal representation [31]. In feature selection, EMTO frameworks partition the global population into collaborative subpopulations to explore complementary search subspaces, including minion-assisted multi-tasking [32], evolutionary multi-task PSO for hyperspectral and gene expression data [33, 34], multi-task bee colony band selection [35], and large-scale multi-objective multi-tasking [36].

Most notably, Zhang et al. [12] introduced Multi-Task Evolutionary Learning (MEL), which establishes a dual-subpopulation PSO framework. In MEL, a population of N particles is partitioned into two collaborative subpopulations, $\mathcal{P}_1$ and $\mathcal{P}_2$ ($|\mathcal{P}_1| = |\mathcal{P}_2| = N/2$), operating on dual complementary search tasks:

Subpopulation 1 ($\mathcal{P}_1$): Continuous swarm exploration. Particles update continuous velocities and positions using PSO dynamics, augmented with an inter-task knowledge transfer channel directing particles toward the historical global best of Subpopulation 2 ($\mathbf{g}_{\text{best},2}$):

$$\mathbf{v}_{1,i}^{(t+1)} = \omega \mathbf{v}_{1,i}^{(t)} + c_1 r_1 (\mathbf{p}_{\text{best},1,i} - \mathbf{p}_{1,i}^{(t)}) + c_2 r_2 (\mathbf{g}_{\text{best},1} - \mathbf{p}_{1,i}^{(t)}) + c_3 r_3 (\mathbf{g}_{\text{best},2} - \mathbf{p}_{1,i}^{(t)}). \tag{8}$$

Subpopulation 2 ($\mathcal{P}_2$): Importance-guided discrete sampling. Rather than continuous mechanics, $\mathcal{P}_2$ samples binary feature masks directly based on a shared feature-importance matrix $W_{\text{MEL}} \in \mathbb{R}^D$.

In MEL, both subpopulations share a scalar fitness function combining classification error and feature subset ratio via static weights:

$$f_{\text{MEL}}(\mathbf{x}) = \alpha \cdot (1 - \text{Acc}(\mathbf{x})) + \beta \cdot \left(\frac{\|\mathbf{x}\|_0}{D} \right), \tag{9}$$

where $\text{Acc}(\mathbf{x})$ is the 5-fold cross-validation accuracy of a 3-NN classifier, and $\alpha = 0.90, \beta = 0.10$ are fixed constants [12].

While MEL demonstrates the utility of cross-task knowledge transfer, its reliance on uniform random initialization, unconstrained importance accumulation without memory discounting, and rigid static weighting creates substantial search inefficiencies in ultra-high-dimensional spaces. The proposed KDS-EL framework directly inherits this dual-subpopulation backbone while resolving these deficiencies through three complementary mechanisms detailed in Section 3.

2.5. Multi-objective feature selection

Feature selection naturally involves two competing criteria: minimizing classification error and minimizing the number of selected features. Multi-Objective Feature Selection (MOFS) algorithms model this trade-off by searching for a Pareto front of non-dominated solutions. Pioneered by Xue et al. [13], diverse multi-objective evolutionary algorithms have been adapted to feature selection, including NSGA-II-FS [37], MOEA/D-FS [38], MOPSO-FS [39], and hybrid filter-wrapper multi-objective architectures [40, 41].

Despite their theoretical appeal, MOFS methods encounter severe operational limitations when scaled to ultra-high-dimensional biomedical data ($D > 10^3$). First, maintaining an external non-dominated archive and executing Pareto ranking over dense high-dimensional vectors incurs substantial computational overhead, often exceeding 110–140 seconds in wall-clock time (Section 4.2.6). Second, under severe sample sparsity, non-dominated sorting tends to produce large, diffused Pareto fronts populated by non-sparse candidate subsets, often retaining 2,000–5,000 active features and failing to deliver the ultra-sparse compact biomarker signatures with fewer than 100 features required for clinical interpretability.

To circumvent Pareto archive bloat and achieve extreme subset sparsity within tight computational budgets, weighted single-objective scalarization remains the preferred formulation in high-dimensional biomarker discovery [12, 16]. However, conventional weighted frameworks enforce fixed static weights throughout search, such as $\mathbf{w} = [0.90, 0.10]$, which fails to adapt to the distinct evolutionary phases of feature selection. KDS-EL overcomes this limitation by introducing a dynamic Two-Phase Weight Scheduling strategy that aggressively eliminates noise features early and fine-tunes classification accuracy late.

2.6. Knowledge-guided search

Integrating domain-specific prior knowledge into meta-heuristics has attracted surging attention as a means to escape the blindness of black-box search [14, 15, 42]. In high-dimensional feature selection, mutual information serves as the primary information-theoretic metric to quantify non-linear feature relevance [30].

Existing knowledge-guided evolutionary algorithms typically adopt one-shot offline injection via either static filter pre-filtering or static velocity biasing. In static filter pre-filtering (e.g., MI-MPODE [15] and NNS-DFPG-CSA [14]), algorithms compute MI rankings offline and initialize the population or restrict the search space to the top-k ranked features. In static velocity biasing (e.g., FWPSO [8]), static feature-weight vectors are incorporated to bias particle velocity equations throughout evolution. While beneficial, one-shot knowledge injection lacks online adaptability: once injected, the static prior is never updated or refined by the empirical trial-and-error feedback generated during subsequent wrapper evaluations. Conversely, learning-based surrogates, such as neural networks [10] or Gaussian processes [43], introduce severe surrogate training overhead and risk model misspecification under sample-scarce conditions.

2.7. Taxonomic comparison

To overcome the collective limitations identified in the existing literature, KDS-EL establishes a closed-loop, knowledge-guided evolutionary learning framework that unifies offline domain priors, online memory harvesting, and stage-aware objective scheduling. Specifically, unlike conventional filter wrappers that permanently eliminate features via hard truncation, KDS-EL uses normalized MI scores as continuous Gaussian-perturbed seeds, preserving the full search space while ensuring an informative, sparse starting basin with $\|\mathbf{x}\|_0 \ll D/2$. In contrast to black-box swarm mutations, KDS-EL dynamically harvests evolutionary fitness feedback into a cumulative importance matrix W, regulated by an exponential forgetting factor $\gamma = 0.99$. The KER operator re-deploys this accumulated knowledge to execute targeted, directional elite flips within a strictly bounded computational budget ($k_e = 3 \ll N$). Furthermore, unlike rigid static weighting with $\mathbf{w} = [0.90, 0.10]$, KDS-EL's Two-Phase Weight Scheduling temporally decouples coarse feature pruning in Phase 1 ($w_f = 0.30$) from fine accuracy refinement in Phase 2 ($w_a = 0.95$), achieving simultaneous gains in accuracy, feature compactness, and runtime efficiency. Table 1 summarizes the structural comparison between KDS-EL and representative state-of-the-art evolutionary feature selection frameworks.

3. The Proposed KDS-EL Method

To overcome the dichotomy between static domain priors and memoryless stochastic search, this section presents the Knowledge-Guided Dual-Phase Swarm Evolutionary Learning (KDS-EL) framework. KDS-EL is designed as a closed-loop evolutionary learning paradigm specifically to overcome three fundamental structural deficiencies of conventional meta-heuristics in ultra-high-dimensional feature selection: (i) the cold-start inefficiency and noise entrapment caused by uniform random initialization in sparse feature spaces, (ii) the permanent loss of cross-iteration feature-importance signals due to memoryless random perturbation, and (iii) the inability of static objective weighting to adapt to the divergent temporal requirements of coarse feature reduction versus fine-grained accuracy refinement.

3.1. The framework of KDS-EL

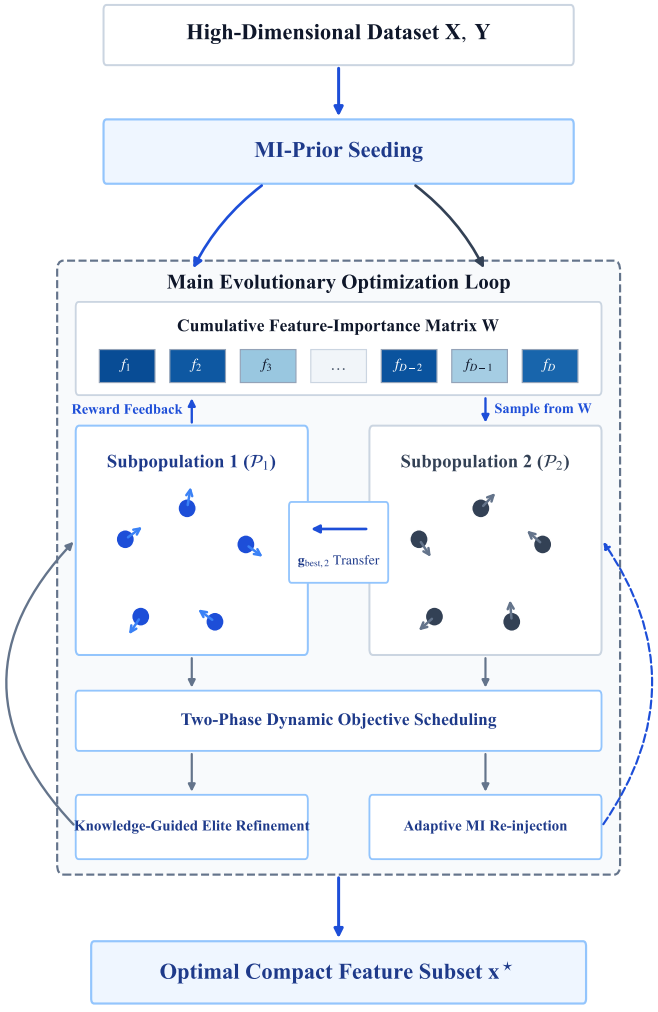


Figure 1: Overall framework of the proposed KDS-EL algorithm.

The macroscopic architecture of KDS-EL is illustrated in Fig. 1. Built upon a cooperative multi-task dual-subpopulation backbone [12], KDS-EL divides a population of $N = 20$ binary particles into two collaborative subpopulations of equal size ($N_1 = N_2 = 10$). Subpopulation 1 ($\mathcal{P}_1$) executes standard continuous-space PSO velocity-position updates with an inter-task knowledge transfer channel directing particles toward the historical global best of Subpopulation 2. Subpopulation 2 ($\mathcal{P}_2$) performs probability-driven feature sampling guided by the population's accumulated feature-importance landscape.

The optimization process operates in a closed loop through four tightly coupled functional stages: (i) *Filter-level quality bootstrapping (offline)*, where mutual information between each feature and class labels is computed once prior to evolution, establishing a normalized prior relevance vector $\mathbf{m} \in [0,1]^D$; (ii) *Knowledge-informed hybrid seeding*, where an initial population of N particles is generated by blending Gaussian-perturbed MI seeds with uniform random particles at a ratio of MI_init_ratio = 0.50, followed by Heaviside step binarization with threshold $\theta = 0.60$; (iii) *Dynamic two-phase evolutionary loop*, where across $T = 100$ iterations, the objective weights $\mathbf{w}(t) = [w_a(t), w_f(t)]$ dynamically transition from aggressive feature pruning ($\mathbf{w}_1 = [0.70, 0.30]$ for $t \le 0.7T$) to accuracy-oriented refinement ($\mathbf{w}_2 = [0.95, 0.05]$ for $t > 0.7T$); and (iv) *Accumulative knowledge harvesting and elite refinement*, where fitness improvements during evaluation are harvested into a continuous feature-importance matrix $W \in \mathbb{R}^D$ discounted by an exponential forgetting factor $\gamma = 0.99$, elite solutions are refined via the KER operator under 1-to-1 in-place replacement, and an adaptive re-initialization mechanism recovers the swarm if search stalls for stall_patience = 20 generations.

3.2. Population initialization strategy

To overcome the cold-start inefficiency and noise entrapment of uniform random initialization in sparse high-dimensional search spaces ($D \gg 10^3$), where standard initialization produces candidate solutions with approximately 50% active features ($\|\mathbf{x}\|_0 \approx D/2$) that waste evaluation budgets in noisy search basins, KDS-EL introduces a filter-wrapper hybrid seeding strategy coupled with adaptive stagnation recovery.

Given the training dataset $\mathcal{D} = \{(\mathbf{f}_i, y_i)\}_{i=1}^{N_{\text{samp}}}$, the mutual information scores $\{I(f_j; Y)\}_{j=1}^{D}$ computed via (6) are normalized into a prior relevance vector $\mathbf{m} \in [0,1]^D$:

$$m_j = \frac{I(f_j;Y) - \min_{1\le k\le D} I(f_k;Y)}{\max_{1\le k\le D} I(f_k;Y) - \min_{1\le k\le D} I(f_k;Y) + \epsilon}, \tag{10}$$

where $\epsilon = 10^{-12}$ prevents division by zero.

To prevent population homogenization while guiding search toward informative dimensions, the initial population is split into two halves based on $\rho = \text{MI_init_ratio} = 0.50$:

$$\mathbf{p}_i = \begin{cases} \text{clip}(\mathbf{m} + \boldsymbol{\varepsilon}_i,\ 0,\ 1), & \boldsymbol{\varepsilon}_i \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_D), \quad \text{for } 1 \le i \le \rho N, \\ \mathbf{u}_i, & \mathbf{u}_i \sim \mathcal{U}(0,1)^D, \quad \text{for } \rho N < i \le N, \end{cases} \tag{11}$$

where $\sigma = 0.10$ represents the Gaussian noise perturbation scale and $\text{clip}(z, 0, 1) = \max(0, \min(1, z))$.

Continuous positions $\mathbf{p}_i = [p_{i,1}, \ldots, p_{i,D}]$ are mapped into binary feature selection vectors $\mathbf{x}_i \in \{0,1\}^D$ using a strict Heaviside step binarization with threshold $\theta = 0.60$:

$$x_{i,j} = \begin{cases} 1, & \text{if } p_{i,j} \ge \theta, \\ 0, & \text{if } p_{i,j} < \theta. \end{cases} \tag{12}$$

Because $\theta = 0.60 > 0.50$, only features possessing above-average mutual information are initialized as active, naturally yielding a sparse initial configuration ($\|\mathbf{x}_i\|_0 \approx 0.15D \sim 0.30D$) that accelerates early convergence, as formulated in Stage 2 of Section 3.6.

3.2.1. Adaptive re-initialization

High-dimensional multimodal search landscapes frequently cause swarm particles to become trapped in suboptimal local attractors. When the global-best fitness f_{gbest} fails to improve for $\tau = \text{stall_patience} = 20$ consecutive iterations, an adaptive re-initialization routine is executed. First, the population is sorted in ascending order of personal-best fitness f_i^{pbest}. The superior half ($\{i \mid \text{rank}(i) \le N/2\}$) is preserved to safeguard historical elite trajectories, whereas the inferior half ($\{i \mid \text{rank}(i) > N/2\}$) is fully refreshed using the Gaussian-perturbed MI distribution:

$$\mathbf{p}_i \leftarrow \text{clip}(\mathbf{m} + \boldsymbol{\varepsilon}_i,\ 0,\ 1), \quad x_{i,j} \leftarrow \mathbb{I}(p_{i,j} \ge \theta). \tag{13}$$

The refreshed particles are then re-evaluated under the current objective weights, and the stagnation counter is reset to zero. This restores diversity within a high-potential subspace without completely discarding established elite knowledge.

3.3. Feature importance learning

To exploit the historical trial-and-error feedback generated across generations and overcome the memoryless amnesia of standard swarm mutations, KDS-EL maintains a continuous cumulative feature-importance matrix $W \in \mathbb{R}^D$, initialized to $\mathbf{0}$.

At generation t, whenever a candidate particle $\mathbf{x}_i^{(t)}$ achieves a strict fitness improvement over its personal best ($f_i^{(t)} < f_i^{\text{pbest}}$), the differential feature subset $C_i^{(t)}$ containing features whose selection status changed is extracted:

$$C_i^{(t)} = \{j \in \{1, \ldots, D\} \mid x_{i,j}^{(t)} \ne p_{\text{best},i,j}\}. \tag{14}$$

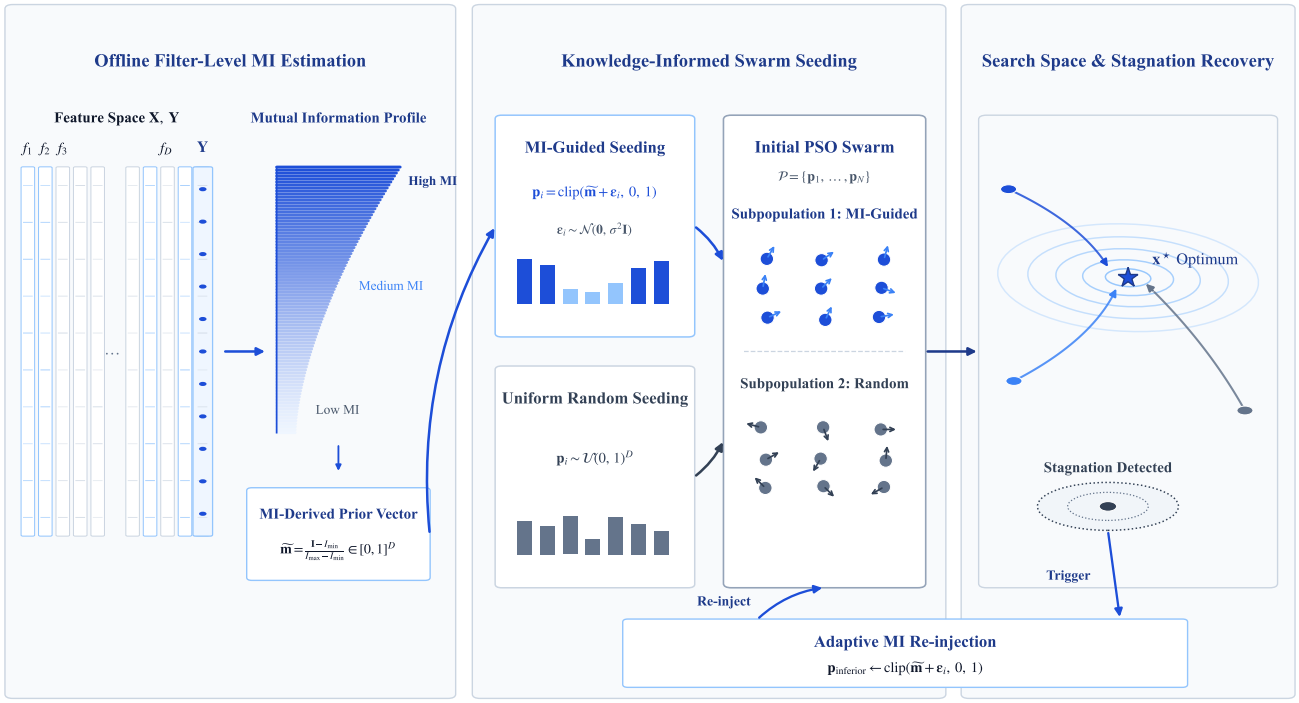


Figure 2: Schematic diagram of the MI-prior seeding and adaptive re-injection mechanism.

Let $\delta_i^{(t)} = f_i^{\text{pbest}} - f_i^{(t)} > 0$ denote the fitness improvement. Each changed feature $j \in C_i^{(t)}$ receives a directional reward proportional to $\delta_i^{(t)}$:

$$W_j \leftarrow W_j + \delta_i^{(t)} \cdot (2x_{i,j}^{(t)} - 1), \quad \forall j \in C_i^{(t)}, \tag{15}$$

where $(2x_{i,j}^{(t)} - 1) = +1$ if feature j was newly activated and -1 if deactivated. Non-improving particles ($\delta_i^{(t)} \le 0$) exert zero impact on W.

To prevent early-discovered features from dominating the matrix indefinitely and allow the algorithm to adapt as the search shifts toward fine-grained subspaces, an exponential memory decay is applied at the conclusion of each generation:

$$W \leftarrow \gamma \cdot W. \tag{16}$$

The decay factor $\gamma = 0.99$ ensures that unless a feature repeatedly contributes to fitness improvements, its score depreciates exponentially, preventing score saturation while dynamically maintaining ranking sensitivity.

3.4. Elite refinement strategy

To exploit the evolutionary knowledge distilled in W and direct local search toward high-potential biomarker dimensions, we propose the Knowledge-Guided Elite Refinement (KER) strategy. Knowledge acquisition in KER requires no additional classifier evaluations, as the importance matrix is updated directly from the fitness comparisons already performed for the population. KER itself spends only a strictly bounded refinement budget ($k_e = 3 \ll N$), preserving the dominant per-iteration computational complexity.

At each generation, the top $k_e = 3$ individuals ranked by fitness are selected. For each elite individual $\mathbf{x}_e \in \{0,1\}^D$, KER executes a probabilistic dual-branch operation (Fig. 3):

Branch 1: Ergodic random exploration ($p_{\text{rand}} = 0.10$): To guarantee search ergodicity, $\lfloor \phi D \rfloor$ features are selected uniformly at random and inverted:

$$x'_{e,j} \leftarrow 1 - x_{e,j}, \quad \forall j \in \mathcal{S}_{\text{rand}}, \quad |\mathcal{S}_{\text{rand}}| = \lfloor \phi D \rfloor, \tag{17}$$

where $\phi = 0.01$ defines the refinement perturbation budget.

Branch 2: Knowledge-guided directional refinement ($1 - p_{\text{rand}} = 0.90$): Each feature $j \in \{1, \ldots, D\}$ is assigned a state-dependent directional flip score σ_j:

$$\sigma_j = (1 - 2x_{e,j}) \cdot W_j = \begin{cases} +W_j, & \text{if } x_{e,j} = 0, \\ -W_j, & \text{if } x_{e,j} = 1. \end{cases} \tag{18}$$

A positive score $+W_j$ favors activating an absent feature with proven positive historical contribution, whereas a negative score $-W_j$ favors deactivating a present feature that previously degraded fitness. We identify the index set $\mathcal{S}_{\text{know}}$ containing the $\lfloor \phi D \rfloor$ features possessing the highest flip scores σ_j:

$$\mathcal{S}_{\text{know}} = \arg \max_{|\mathcal{S}| = \lfloor \phi D \rfloor} \sum_{j \in \mathcal{S}} \sigma_j, \quad \mathcal{S} \subseteq \{1, \ldots, D\}, \tag{19}$$

and invert the selected features: $x'_{e,j} \leftarrow 1 - x_{e,j}, \forall j \in \mathcal{S}_{\text{know}}$.

Greedy 1-to-1 in-place replacement: The refined mutant $\mathbf{x}'_e$ is evaluated in a closed loop, and admitted via greedy 1-to-1 in-place replacement:

$$\mathbf{x}_e^{\star} = \begin{cases} \mathbf{x}'_e, & \text{if } f(\mathbf{x}'_e, \mathbf{w}(t)) < f(\mathbf{x}_e, \mathbf{w}(t)), \\ \mathbf{x}_e, & \text{otherwise.} \end{cases} \tag{20}$$

If improved, $\mathbf{x}'_e$ immediately overwrites $\mathbf{x}_e$ in the population and updates the global best if $f(\mathbf{x}'_e) < f_{\text{gbest}}$. Because elite_k = 3 is fixed, KER executes within an exact bounded overhead of $k_e = 3$ evaluations per generation. The detailed execution of KER is formalized in Algorithm 1.

3.5. Two-phase weight scheduling

To align objective weights with the temporal structure of feature selection and resolve the conflict between early dimensionality reduction and late accuracy refinement, KDS-EL introduces the Two-Phase Weight Scheduling (TPS) strategy. The total generational budget $T = 100$ is partitioned at split point $\kappa = 0.70$. The dynamic dual-objective weight vector $\mathbf{w}(t) = [w_a(t), w_f(t)]$ is governed by:

$$\mathbf{w}(t) = \begin{cases} [0.70,\ 0.30], & \text{for } 1 \le t \le \lfloor \kappa T \rfloor, \\ [0.95,\ 0.05], & \text{for } \lfloor \kappa T \rfloor < t \le T. \end{cases} \tag{21}$$

The unified scalar fitness $f(\mathbf{x}, \mathbf{w}(t))$ of a binary feature subset $\mathbf{x} \in \{0,1\}^D$ at generation t is formulated as:

$$f(\mathbf{x}, \mathbf{w}(t)) = w_a(t) \cdot (1 - \text{Acc}(\mathbf{x})) + w_f(t) \cdot \left(\frac{\|\mathbf{x}\|_0}{D} \right), \tag{22}$$

where $\text{Acc}(\mathbf{x})$ denotes 5-fold cross-validation classification accuracy evaluated by a 3-NN classifier, and $\|\mathbf{x}\|_0/D$ represents the normalized feature selection ratio.

In Phase 1 ($t \le 0.7T$), the heavy feature-ratio penalty ($w_f = 0.30$) drives rapid elimination of non-informative dimensions. In Phase 2 ($t > 0.7T$), the weight switches to prioritize classification accuracy ($w_a = 0.95$), enabling fine-tuning of delicate biomarker interactions over the surviving compact subspace.

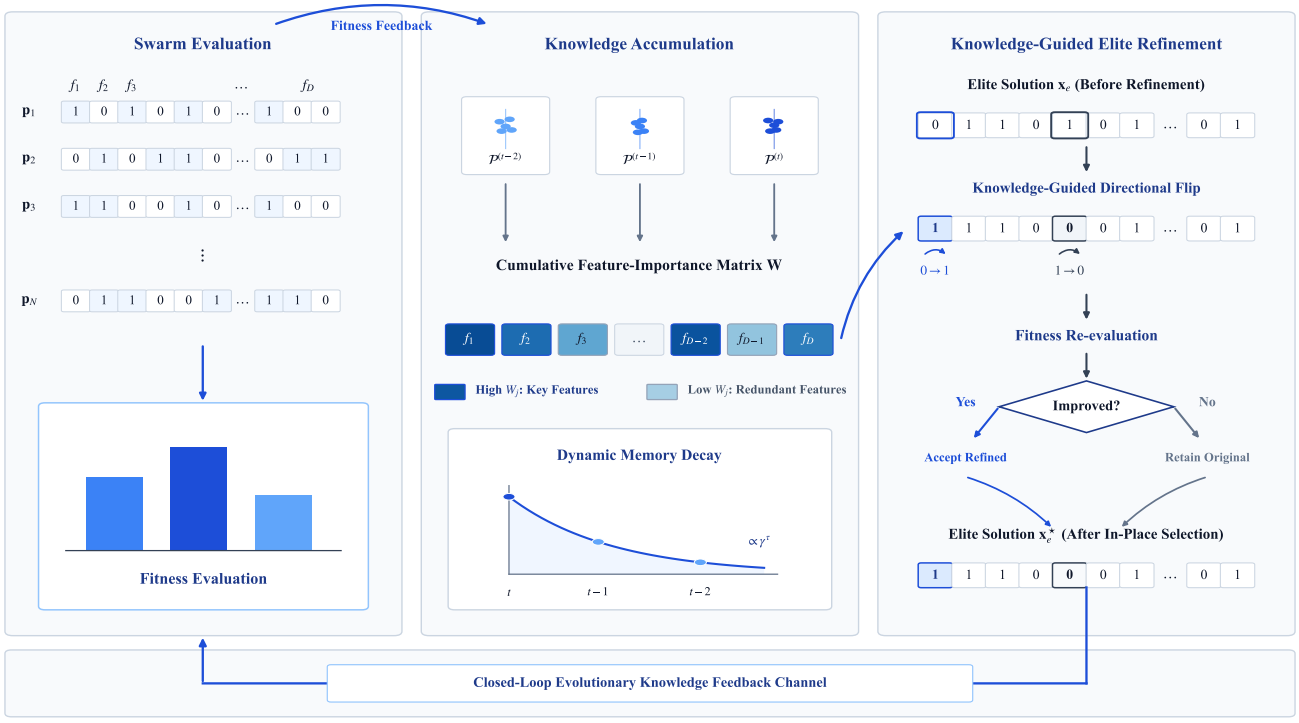


Figure 3: Flowchart of the KER strategy.

Algorithm 1 Knowledge-guided elite refinement (KER)

Require: Elite candidate solution $\mathbf{x}_e \in \{0,1\}^D$, cumulative importance matrix $W \in \mathbb{R}^D$, perturbation budget $\phi \in (0,1)$, exploration probability $p_{\text{rand}} \in [0,1]$, dynamic objective weights $\mathbf{w}(t)$
Ensure: Refined elite candidate $\mathbf{x}_e^\star \in \{0,1\}^D$

$k_{\text{flip}} \leftarrow \max(1, \lfloor \phi \cdot D \rfloor)$ ▷ Determine perturbation budget
Sample random variable $r \sim \mathcal{U}(0,1)$
if $r < p_{\text{rand}}$ **then** ▷ Branch 1: Ergodic random exploration
  Sample random feature subset $\mathcal{S}_{\text{rand}} \subset \{1,\dots,D\}$ with $|\mathcal{S}_{\text{rand}}| = k_{\text{flip}}$
  $\mathbf{x}'_e \leftarrow \mathbf{x}_e$; $x'_{e,j} \leftarrow 1 - x_{e,j}$, $\forall j \in \mathcal{S}_{\text{rand}}$
else ▷ Branch 2: Knowledge-guided directional refinement
  for $j = 1$ **to** D **do**
    $\sigma_j \leftarrow (1 - 2x_{e,j}) \cdot W_j$ ▷ Compute directional flip score
  end for
  Identify subset $\mathcal{S}_{\text{know}}$ of size k_{flip} with top σ_j scores via (19)
  $\mathbf{x}'_e \leftarrow \mathbf{x}_e$; $x'_{e,j} \leftarrow 1 - x_{e,j}$, $\forall j \in \mathcal{S}_{\text{know}}$
end if
Evaluate offspring fitness $f' \leftarrow f(\mathbf{x}'_e, \mathbf{w}(t))$ using (22)
if $f' < f(\mathbf{x}_e, \mathbf{w}(t))$ **then**
  $\mathbf{x}_e^\star \leftarrow \mathbf{x}'_e$ ▷ Greedy 1-to-1 in-place replacement
else
  $\mathbf{x}_e^\star \leftarrow \mathbf{x}_e$ ▷ Retain original elite
end if
return $\mathbf{x}_e^\star$

Algorithm 2 The framework of the proposed KDS-EL algorithm

Require: Training dataset $\mathcal{D} = (X, Y)$, population size N, max iterations T, binarization threshold θ, MI seeding ratio ρ, decay factor γ, elite pool size k_e, perturbation fraction ϕ, random branch probability p_{rand}, phase split ratio κ, stagnation threshold τ
Ensure: Optimal compact feature subset $\mathbf{x}^\star \in \{0,1\}^D$ and optimal fitness f_{gbest}

// Stage 1: Filter-level relevance extraction
Compute and normalize mutual information vector $\mathbf{m} \in [0,1]^D$ via (6) and (10)
// Stage 2: Knowledge-informed hybrid population seeding
Initialize population $\mathcal{P} = \{\mathbf{x}_i\}_{i=1}^N$ with ρN Gaussian MI seeds and $(1-\rho)N$ uniform particles via (11)–(12)
Initialize cumulative feature-importance matrix $W \leftarrow \mathbf{0} \in \mathbb{R}^D$, stall_count $\leftarrow 0$, and initial weights $\mathbf{w}(1) = [0.70, 0.30]$
Evaluate initial population fitness $\{f_i\}_{i=1}^N$ via (22) and initialize personal/global bests $\mathbf{p}_{\text{best},i}$, $\mathbf{g}_{\text{best}}$
// Stage 3: Dynamic two-phase evolutionary optimization loop
for $t = 1$ **to** T **do**
  Update dynamic objective weight vector $\mathbf{w}(t)$ according to (21)
  Update Subpopulation 1 ($\mathcal{P}_1$) via continuous PSO with inter-task knowledge transfer channel [12]
  Update Subpopulation 2 ($\mathcal{P}_2$) via probability-driven feature sampling guided by activation vector $\mathbf{P} = \sigma(W)$
  Evaluate candidate solution fitness $\{f(\mathbf{x}_i, \mathbf{w}(t))\}_{i=1}^N$ using 5-fold cross-validation k-NN classifier via (22)
  Update cumulative matrix W with differential fitness improvements and apply exponential memory decay γ via (15)–(16)
  Update personal bests $\mathbf{p}_{\text{best},i}$ and identify top k_e elite candidate indices $\mathcal{E}$
  // Knowledge-guided elite refinement (KER)
  for each $e \in \mathcal{E}$ **do**
    $\mathbf{p}_{\text{best},e} \leftarrow \text{KER}(\mathbf{p}_{\text{best},e}, W, \phi, p_{\text{rand}}, \mathbf{w}(t))$ ▷ Call Algorithm 1
  end for
  // Global best tracking and adaptive stagnation recovery
  Update global best $\mathbf{g}_{\text{best}} \leftarrow \arg\min_i f_i^{\text{pbest}}$ and update stall_count
  if stall_count $\geq \tau$ **then**
    Refresh inferior half of swarm using Gaussian-perturbed MI distribution via (13)
    stall_count $\leftarrow 0$
  end if
end for
return Optimal feature subset $\mathbf{x}^\star \leftarrow \mathbf{g}_{\text{best}}$ and optimal fitness f_{gbest}

3.6. Main algorithm

Algorithm 2 consolidates the holistic execution workflow of KDS-EL into a modular multi-stage optimization pipeline. The framework coordinates offline filter relevance extraction, hybrid stochastic seeding, two-phase dynamic objective scheduling, collaborative dual-subpopulation search, differential knowledge matrix harvesting, and targeted KER elite refinement.

3.7. Complexity analysis

To establish the computational efficiency of KDS-EL, we conduct a rigorous asymptotic complexity analysis covering all algorithmic components.

3.7.1. Time complexity decomposition

The execution time of KDS-EL is decomposed into two distinct phases:

Offline preprocessing phase (mutual information calculation): Computing the mutual information $I(f_j; Y)$ between each feature f_j and class label Y requires $O(N_{\text{samp}})$ operations per feature, where N_{samp} is the number of training samples. For all D features, computing and sorting the prior MI vector incurs:

$$T_{\text{pre}} = O(D \cdot N_{\text{samp}} + D \log D). \tag{23}$$

Because this step is executed only once prior to the evolutionary loop, its overhead is entirely negligible relative to iterative wrapper evaluations.

Iterative evolutionary main loop (T iterations): In each of the T generations, computational costs arise from: (i) updating particle continuous positions ($O(N \cdot D)$), (ii) differential matrix updating and decay ($O(N \cdot D + D)$), (iii) refining k_e elites via KER ($O(k_e \cdot (D + T_{\text{eval_single}}))$), (iv) occasional adaptive reinitialization ($O(\frac{N}{2} \cdot (D + T_{\text{eval_single}}))$), and (v) evaluating 5-fold cross-validation k-NN classification accuracy for N candidate particles:

$$T_{\text{eval}} = O(N \cdot N_{\text{samp}}^2 \cdot d_{\text{selected}}), \tag{24}$$

where $d_{\text{selected}} = \|\mathbf{x}\|_0$ is the number of active features in candidate $\mathbf{x}$.

Summing these terms across T generations yields the total asymptotic time complexity:

$$\mathcal{T}_{\text{total}} = O\Big(D \log D + T \cdot [N \cdot D + (N + k_e) \cdot N_{\text{samp}}^2 \cdot d_{\text{selected}}]\Big). \tag{25}$$

3.7.2. Theoretical complexity equivalence and practical acceleration

Because $k_e = 3 \ll N = 20$ and $N_{\text{samp}}^2 \cdot d_{\text{selected}} \gg D$, the dominant asymptotic per-generation complexity of KDS-EL is strictly governed by wrapper evaluations:

$$\mathcal{T}_{\text{KDS-EL}} = O(T \cdot N \cdot N_{\text{samp}}^2 \cdot d_{\text{selected}}), \tag{26}$$

which is identically of the same asymptotic order as standard MEL [12] ($O(T \cdot N \cdot N_{\text{samp}}^2 \cdot d_{\text{MEL}})$).

Crucially, in standard MEL, random initialization and sluggish feature reduction result in dense candidate subsets ($d_{\text{MEL}} \approx 2{,}722$ active features on average). In contrast, KDS-EL's MI seeding and Phase 1 aggressive feature pruning rapidly compress the active feature count to $d_{\text{KDS-EL}} \approx 88$ features. Because k-NN distance calculation scales linearly with d_{selected}, evaluating a KDS-EL particle is over an order of magnitude faster than evaluating an unpruned MEL particle. Consequently, KDS-EL delivers a 3.4× empirical wall-clock speedup, reducing average runtime from 175.0 s for MEL to 51.4 s while incorporating richer knowledge operators.

3.7.3. Space complexity

The memory footprint of KDS-EL comprises: (i) the population continuous and binary position matrices ($O(N \cdot D)$), (ii) personal-best vectors ($O(N \cdot D)$), (iii) the cumulative feature-importance matrix W ($O(D)$), and (iv) the precomputed normalized MI vector $\mathbf{m}$ ($O(D)$). The overall space complexity is:

$$\mathcal{S}_{\text{total}} = O(N \cdot D + N_{\text{samp}} \cdot D), \tag{27}$$

which scales strictly linearly with dimension D, ensuring minimal memory consumption.

4. Experimental results and discussion

This section outlines the experimental framework and presents a comprehensive empirical evaluation of KDS-EL across eight analytical dimensions: classification accuracy, feature subset compactness, fitness convergence dynamics, statistical significance, performance progression before and after algorithmic improvement, computational efficiency, hyperparameter sensitivity, and in-depth discussion on biomedical ceiling effects. All aggregate numbers are computed over the 20 benchmark datasets described in Section 4.1.

4.1. Experimental setup

This subsection outlines the experimental framework designed to evaluate the classification performance, dimensionality reduction capability, computational efficiency, and hyperparameter robustness of KDS-EL.

4.1.1. High-dimensional benchmark datasets

To validate the performance and generalization ability of KDS-EL across diverse problem domains, a total of 20 benchmark datasets widely adopted in the evolutionary feature selection literature [8, 11, 12] are selected. These datasets encompass diverse categories and sample sizes across biomedical gene-expression microarrays, text categorization corpora, image recognition, speech processing, and synthetic benchmarks. Such diverse data distributions and dimensional complexities effectively evaluate the generalization capability and feature reduction efficacy of feature selection algorithms under challenging conditions. The detailed characteristics of the 20 benchmark datasets are summarized in Table 2.

4.1.2. Comparison algorithms

We select two primary categories of comparison algorithms for analysis with the proposed KDS-EL framework, as summarized in Table 1. The first category comprises specialized single-objective evolutionary algorithms in the field of high-dimensional feature selection, which includes: feature-weight particle swarm optimization (FWPSO) [8], particle swarm optimization with multi-criterion scoring (PSO-CSM) [11], self-adaptive differential evolution with adaptive weighting (SaWDE) [9], information gain-guided particle swarm optimization (IGPSO) [10], and binary particle swarm optimization (BPSO) [7]. The relevant parameter settings are shown in Table 3.

Another category of representative algorithms encompasses multi-objective and multi-task feature selection approaches, including non-dominated sorting genetic algorithm II (NSGA-II-FS) [37], multi-objective evolutionary algorithm based on decomposition (MOEA/D-FS) [38], multi-objective particle swarm optimization (MOPSO-FS) [39], fuzzy multi-objective particle swarm optimization (fMOPSO-FS) [44], wrapper–filter hybrid multi-objective PSO (W-MOPSO-FS), and multi-task evolutionary learning (MEL) [12]. In addition, two constituent variants of KDS-EL are investigated in Section 4.2.5 to evaluate individual component contributions.

4.1.3. Evaluation metrics and statistical test protocols

Algorithm performance is evaluated across three primary criteria: classification accuracy, the number of selected features alongside the feature selection ratio, and computational runtime.

To determine whether the empirical improvements of KDS-EL are statistically significant, we follow the multi-problem statistical testing framework recommended by Demšar [45]. Specifically, the non-parametric Friedman test [46] is utilized to establish global algorithm rankings across all 20 benchmark datasets. Pairwise Wilcoxon signed-rank tests at the $\alpha = 0.05$ significance level assess directional superiority between KDS-EL and each competitor, with the Holm–Bonferroni step-down procedure [47] applied to rigorously control the family-wise error rate across multiple comparisons.

Table 2: Characteristics of the 20 benchmark datasets.

No.	Dataset	Samples (n)	Features (D)	Classes (C)	Domain
1	Adenoma	36	7,457	2	Gene expression
2	ALL_AML	72	7,129	2	Gene expression
3	ALL1	128	12,625	2	Gene expression
4	ALL4	93	12,625	2	Gene expression
5	Colon	62	2,000	2	Gene expression
6	DLBCL	77	7,129	2	Gene expression
7	Gastric	65	22,645	2	Gene expression
8	Gastric1	144	22,283	2	Gene expression
9	Gastric2	124	22,283	2	Gene expression
10	Leukaemia	72	7,129	2	Gene expression
11	Lymphoma	45	4,026	2	Gene expression
12	Myeloma	173	12,625	2	Gene expression
13	Prostate	102	12,625	2	Gene expression
14	SRBCT	83	2,308	4	Gene expression
15	T1D	101	54,675	2	Gene expression
16	BASEHOCK	1,993	4,862	2	Text
17	PCMAC	1,943	3,289	2	Text
18	COIL20	1,440	1,024	20	Image
19	ISOLET	1,560	617	26	Speech
20	madelon	2,600	500	2	Synthetic

4.2. Experimental analysis

This subsection presents a comprehensive empirical evaluation of KDS-EL across eight complementary analytical dimensions: classification accuracy, feature subset compactness, fitness convergence dynamics, statistical significance, performance progression before and after algorithmic improvement, computational efficiency, hyperparameter sensitivity, and in-depth discussion on biomedical ceiling effects.

4.2.1. Comparison of classification accuracy

The KDS-EL framework was tested with 13 comparison algorithms on 20 benchmark datasets with high-dimensional feature selection. Table 4 presents

Table 3: Relevant parameter settings for all algorithms.

Algorithm / Component	Parameter setting
Common experimental parameters:	
Population size N	20
Maximum generations $T_{\max}$	100
Classifier	3-NN with 5-fold stratified CV
Binarization threshold θ	0.60
Independent runs	10
Proposed KDS-EL framework:	
Subpopulations	$N_1 = 10,\ N_2 = 10$
Inertia weight ω	0.90
Acceleration coefficients c_1, c_2, c_3	2.0, 2.0, 2.0
Knowledge decay factor γ	0.99
Perturbation fraction ϕ	0.01
Elite pool size k_e	3
Random exploration rate p_{rand}	0.10
MI initialization ratio ρ	0.50
Stagnation threshold τ	20 generations
Phase split ratio κ	0.70
Phase-1 weights $\mathbf{w}_1$	$[0.70,\ 0.30]$
Phase-2 weights $\mathbf{w}_2$	$[0.95,\ 0.05]$
Ablative variants:	
KDS-EL w/o TPS	Static weights $\mathbf{w}_s = [0.90,\ 0.10]$
KDS-EL w/o TPS&KER	Without KER and TPS, $\mathbf{w}_s = [0.90,\ 0.10]$
Original MEL [12]	Uniform init, $\mathbf{w}_s = [0.90,\ 0.10]$
Competitor algorithms:	
FWPSO [8]	$\omega \in [0.4, 0.9],\ c_1 = c_2 = 2.0,\ \alpha_{\text{sa}} = 0.95$
PSO-CSM [11]	$\omega = 0.7298,\ c_1 = c_2 = 1.49618,\ p_m = 0.05$
SaWDE [9]	$F \in [0.1, 0.9],\ CR \in [0.1, 0.9]$
IGPSO [10]	$\omega \in [0.4, 0.9],\ c_1 = c_2 = 2.0$, IG scoring
BPSO [7]	$\omega \in [0.4, 0.9],\ c_1 = c_2 = 2.0,\ V_{\max} = 6.0$
NSGA-II-FS [37]	$p_c = 0.9,\ \eta_c = 20,\ p_m = 1/D,\ \eta_m = 20$
MOEA/D-FS [38]	Neighborhood $T_n = 5$, Tchebycheff
MOPSO-FS [39]	Archive 100, grid inflation $\alpha_g = 0.1$
fMOPSO-FS [44]	Archive 100, $\omega = 0.7298,\ c_{1,2} = 1.49618$
W-MOPSO-FS	Archive 100, dynamic filter–wrapper

the results of the experiments for average classification accuracy and standard deviation, with the optimal values in bold.

First, comparing the KDS-EL algorithm to the 13 comparison algorithms overall, KDS-EL achieved the highest average classification accuracy of 0.9574 across the benchmark suite, securing the highest score on 15 of the 20 high-dimensional datasets. Among them, on ultra-high-dimensional microarrays with severe sample sparsity, KDS-EL delivers substantial performance advantages, reaching classification accuracies of 0.9522 on the 12,625-dimensional *ALL4*, 0.9500 on *Colon*, and 0.8598 on the 54,675-dimensional *T1D*, consistently establishing distinct margins over competing single- and multi-objective algorithms. Secondly, compared to established single-objective evolutionary algorithms such as FWPSO, PSO-CSM, SaWDE, IGPSO, and BPSO, KDS-EL improves overall mean accuracy from the 0.8698–0.9177 range to 0.9574, indicating that closed-loop knowledge transfer and Two-Phase Weight Scheduling effectively overcome the blind stochastic trials of standard swarm operators. Compared with multi-objective methods including W-MOPSO-FS, fMOPSO-FS, NSGA-II-FS, MOEA-D-FS, and MOPSO-FS, KDS-EL maintains a consistent advantage of over 0.04 to 0.05 in mean classification score. Furthermore, compared with the original MEL algorithm, KDS-EL achieves an accuracy improvement of 0.0706, which indicates that mutual information initialization and knowledge-guided elite refinement effectively enhance the algorithm's biomarker mining capability and local exploitation. Overall, the KDS-EL algorithm searches for a superior subset of features and obtains higher classification accuracy.

A closer examination of the individual datasets and algorithmic dispersion provides further insights into the search behavior and robustness of KDS-EL. On the few moderate-dimensional datasets where KDS-EL does not record the nominal highest accuracy, such as *BASEHOCK*, *COIL20*, and *ISOLET*, competitors such as NSGA-II-FS achieve marginally higher accuracy by 0.003 to 0.020 only by retaining massive subsets of 185 to 2,073 features, whereas KDS-EL attains competitive accuracy with only 60 to 134 features, achieving an order-of-magnitude reduction in feature redundancy. On near-perfect saturated datasets such as *ALL1* and *SRBCT*, KDS-EL attains 0.9998 and 0.9986 accuracy with minimal signatures of 39 and 22 features. Furthermore, KDS-EL demonstrates remarkably low run-to-run dispersion, exhibiting a standard deviation of only 0.0089 on *Gastric1*, compared with 0.0215 for FWPSO and 0.0381 for MEL. This confirms that hybrid MI initialization and knowledge-guided elite refinement not only deliver higher accuracy in finding optimal biomarker subsets, but also substantially enhance the stability and robustness of the algorithm.

4.2.2. Comparison of the number of selected features

The KDS-EL framework was evaluated against 13 comparison algorithms across 20 high-dimensional benchmark datasets. Table 5 presents the experimental results for the average number of selected features and the feature selection ratio, with the optimal values in bold.

First, comparing the KDS-EL algorithm to the 13 comparison algorithms overall, KDS-EL achieved the most compact feature subset on all 20 benchmark datasets, selecting an overall mean of only 87.8 features, corresponding to approximately 2%–3% of the original feature space. The feature reduction capability of KDS-EL is particularly evident on ultra-high-dimensional datasets such as *ALL1*, *Colon*, *Gastric1*, and *Prostate*, where KDS-EL extracts minimal biomarker signatures of merely 39, 14, 31, and 26 features while achieving superior classification accuracies between 0.9500 and 0.9998. Secondly, compared to the single-objective evolutionary algorithms, even the most compact competitor, FWPSO, which retains 1,068.9 features on average, selects over 12 times more features than KDS-EL, indicating that two-phase weight scheduling effectively enforces aggressive dimensionality reduction in the early evolutionary phase. Compared with the original MEL algorithm, which retains 2,722.2 features, KDS-EL achieves a 31-fold reduction in feature subset size, which indicates that mutual information initialization and knowledge-guided elite refinement effectively steer the swarm toward sparse, informative subspaces rather than dense redundant regions. Furthermore, although multi-objective algorithms such as NSGA-II-FS and fMOPSO-FS obtain competitive accuracy on certain datasets, they universally retain extensive subsets ranging from 2,960.0 to 5,263.3 features, failing to deliver the ultra-sparse biomarker signatures required for biomedical interpretability. Overall, the KDS-EL algorithm searches for a significantly more compact subset of features while maintaining higher classification accuracy.

From the perspective of algorithmic components, the mean feature count exhibits a strictly monotonic reduction across the ablative progression: from 2,722.2 for MEL, down to 290.5 for KDS-EL w/o TPS&KER, 121.7 for KDS-EL w/o TPS, and finally 87.8 for KDS-EL. This progressive compression demonstrates that each proposed strategy plays an indispensable role in eliminating non-informative and redundant dimensions: MI initialization provides an informative sparse starting basin, KER directs targeted elite flips toward high-potential biomarker dimensions, and TPS dynamically decouples coarse feature pruning from fine accuracy refinement. Consequently, KDS-EL establishes clear Pareto dominance over existing evolutionary feature selection methods by simultaneously maximizing classification performance and minimizing feature redundancy.

4.2.3. Fitness value and convergence analysis

The fitness value represents the algorithm's comprehensive trade-off between classification accuracy and the number of selected features during search, which directly reflects the algorithm's global optimization capability. In this subsection, we present the convergence trajectories of the fitness values across the 20 benchmark datasets for KDS-EL and its ablative variants over 100 generations, as illustrated in Fig. 4, depicting the overall mean trajectory alongside eight representative datasets spanning diverse dimensionalities.

As demonstrated in Fig. 4, the KDS-EL framework consistently achieves the lowest fitness values across all benchmark datasets, indicating its superior global search capabilities in complex, ultra-high-dimensional spaces. In the early evolutionary stages, mutual information initialization rapidly seeds the swarm in informative low-dimensional basins, allowing KDS-EL to attain high fitness within the first 10 iterations, whereas the unguided MEL baseline requires 40 to 60 iterations to escape inferior search regions. Although the initial dual-subpopulation division and aggressive feature-size penalty restrict early accuracy exploitation, the collaborative knowledge transfer between subpopulations continuously enhances population diversity and prevents premature stagnation. Consequently, KDS-EL achieves substantially lower fitness values than its constituent variants and the original MEL algorithm across the entire evolutionary process.

The convergence curves depicted in Fig. 4 clearly illustrate the dynamic optimization advantages of KDS-EL across distinct evolutionary phases. During the intermediate search phase between generations 15 and 70, Phase-1 objective weights aggressively eliminate redundant dimensions, sustaining a steep descent in fitness value. Upon entering Phase 2 at generation 70, the transition to accuracy-prioritized weights refocuses computational effort on fine-tuning

Table 4: Comparison of classification accuracy (mean ± std) across 20 benchmark datasets.

Dataset	KDS-EL	w/o TPS	w/o KER	MEL	FWPSO	PSO-CSM	SaWDE	IGPSO	BPSO	W-MPSO	fMPSO	NSGA-II	MOEA/D	MOPSO
ALL1	0.9998±0.0001	0.9996±0.0000^{+}	0.9996±0.0000^{+}	0.9664±0.0023^{+}	0.9952±0.0015^{+}	0.9900±0.0002^{+}	0.9710±0.0011^{+}	0.9802±0.0002^{+}	0.9534±0.0003^{+}	**1.0000±0.0000**$^{-}$	**1.0000±0.0000**$^{-}$	**1.0000±0.0000**$^{-}$	0.9985±0.0031$^{\approx}$	**1.0000±0.0000**$^{-}$
ALL4	**0.9522±0.0061**	0.9418±0.0113$^{\approx}$	0.9415±0.0112^{+}	0.8704±0.0661^{+}	0.8617±0.0304^{+}	0.8363±0.0135^{+}	0.8062±0.0217^{+}	0.7914±0.0179^{+}	0.7569±0.0069^{+}	0.8068±0.0047^{+}	0.8015±0.0089^{+}	0.7694±0.0118^{+}	0.7737±0.0184^{+}	0.7654±0.0074^{+}
ALL_AML	**0.9995±0.0002**	0.9990±0.0002^{+}	0.9886±0.0016^{+}	0.9539±0.0008^{+}	0.9850±0.0067^{+}	0.9748±0.0039^{+}	0.9591±0.0012^{+}	0.9669±0.0003^{+}	0.9426±0.0034^{+}	0.9943±0.0070$^{\approx}$	0.9900±0.0065^{+}	0.9857±0.0000^{+}	0.9857±0.0000^{+}	0.9857±0.0000^{+}
Adenoma	**0.9996±0.0002**	**0.9996±0.0001**$^{\approx}$	0.9958±0.0081^{+}	0.9410±0.0011^{+}	0.9783±0.0143^{+}	0.9665±0.0108^{+}	0.9461±0.0014^{+}	0.9527±0.0003^{+}	0.9287±0.0002^{+}	0.9800±0.0131^{+}	0.9743±0.0086^{+}	0.9714±0.0000^{+}	0.9714±0.0000^{+}	0.9714±0.0000^{+}
BASEHOCK	0.9113±0.0120	0.8997±0.0114^{+}	0.8597±0.0062^{+}	0.8518±0.0063^{+}	0.8715±0.0095^{+}	0.8637±0.0050^{+}	0.8860±0.0065^{+}	0.8479±0.0062^{+}	0.8597±0.0036^{+}	0.8847±0.0066^{+}	0.8861±0.0045^{+}	**0.9238±0.0082**$^{-}$	0.9150±0.0086$^{\approx}$	0.8920±0.0034^{+}
COIL20	0.9939±0.0011	0.9919±0.0008^{+}	0.9619±0.0007^{+}	0.9634±0.0017^{+}	0.9894±0.0018^{+}	0.9782±0.0006^{+}	0.9763±0.0017^{+}	0.9707±0.0010^{+}	0.9573±0.0006^{+}	0.9960±0.0009^{-}	0.9960±0.0005^{-}	**0.9971±0.0007**$^{-}$	0.9958±0.0010^{-}	0.9964±0.0009^{-}
Colon	**0.9500±0.0075**	0.9347±0.0094^{+}	0.9211±0.0092^{+}	0.8548±0.0276^{+}	0.9221±0.0203^{+}	0.8880±0.0090^{+}	0.8867±0.0110^{+}	0.8636±0.0101^{+}	0.8529±0.0059^{+}	0.9033±0.0072^{+}	0.8988±0.0071^{+}	0.8731±0.0097^{+}	0.8745±0.0145^{+}	0.8806±0.0104^{+}
DLBCL	**0.9995±0.0002**	0.9993±0.0004$^{\approx}$	0.9902±0.0016^{+}	0.9483±0.0058^{+}	0.9819±0.0092^{+}	0.9819±0.0048^{+}	0.9629±0.0055^{+}	0.9655±0.0063^{+}	0.9435±0.0069^{+}	0.9974±0.0052$^{\approx}$	0.9960±0.0061$^{\approx}$	0.9781±0.0060^{+}	0.9781±0.0060^{+}	0.9872±0.0058^{+}
Gastric	**0.9782±0.0034**	0.9677±0.0093^{+}	0.9748±0.0088$^{\approx}$	0.8955±0.0786^{+}	0.9191±0.0340^{+}	0.8874±0.0087^{+}	0.8234±0.0191^{+}	0.8564±0.0087^{+}	0.8046±0.0093^{+}	0.8908±0.0046^{+}	0.8846±0.0077^{+}	0.8154±0.0069^{+}	0.8169±0.0083^{+}	0.8215±0.0102^{+}
Gastric1	**0.9800±0.0003**	0.9738±0.0004^{+}	0.9642±0.0025^{+}	0.9310±0.0155^{+}	0.9597±0.0032^{+}	0.9496±0.0024^{+}	0.9348±0.0030^{+}	0.9402±0.0018^{+}	0.9145±0.0002^{+}	0.9588±0.0021^{+}	0.9581±0.0000^{+}	0.9588±0.0021^{+}	0.9554±0.0034^{+}	0.9588±0.0021^{+}
Gastric2	**0.9981±0.0029**	0.9936±0.0029^{+}	0.9872±0.0038^{+}	0.9522±0.0170^{+}	0.9837±0.0049^{+}	0.9710±0.0002^{+}	0.9532±0.0038^{+}	0.9623±0.0001^{+}	0.9357±0.0034^{+}	0.9837±0.0000^{+}	0.9838±0.0002^{+}	0.9765±0.0025^{+}	0.9765±0.0025^{+}	0.9832±0.0025^{+}
ISOLET	0.8894±0.0056	0.8898±0.0090$^{\approx}$	0.8557±0.0063^{+}	0.8501±0.0067^{+}	0.8822±0.0109$^{\approx}$	0.8584±0.0043^{+}	0.8903±0.0076$^{\approx}$	0.8468±0.0070^{+}	0.8607±0.0027^{+}	0.8883±0.0032$^{\approx}$	0.8874±0.0066$^{\approx}$	**0.9090±0.0059**$^{-}$	0.8963±0.0059^{-}	0.8828±0.0036^{+}
Leukaemia	**0.9995±0.0002**	0.9993±0.0002$^{\approx}$	0.9895±0.0017^{+}	0.9534±0.0015^{+}	0.9886±0.0059^{+}	0.9775±0.0059^{+}	0.9587±0.0017^{+}	0.9682±0.0037^{+}	0.9416±0.0002^{+}	0.9886±0.0057$^{\approx}$	0.9886±0.0057^{+}	0.9857±0.0000^{+}	0.9857±0.0000^{+}	0.9857±0.0000^{+}
Lymphoma	**0.9994±0.0003**	**0.9994±0.0004**$^{\approx}$	0.9868±0.0102^{+}	0.8731±0.0239^{+}	0.9404±0.0306^{+}	0.9206±0.0098^{+}	0.8841±0.0092^{+}	0.8861±0.0139^{+}	0.8585±0.0081^{+}	0.9178±0.0102^{+}	0.9200±0.0109^{+}	0.8844±0.0133^{+}	0.8867±0.0067^{+}	0.8911±0.0067^{+}
Myeloma	**0.9155±0.0101**	0.8971±0.0072^{+}	0.8825±0.0084^{+}	0.8450±0.0244^{+}	0.8702±0.0069^{+}	0.8550±0.0079^{+}	0.8522±0.0071^{+}	0.8304±0.0077^{+}	0.8186±0.0056^{+}	0.8535±0.0037^{+}	0.8529±0.0060^{+}	0.8594±0.0070^{+}	0.8482±0.0115^{+}	0.8425±0.0035^{+}
PCMAC	**0.8705±0.0139**	0.8424±0.0163^{+}	0.8007±0.0051^{+}	0.7870±0.0080^{+}	0.8103±0.0090^{+}	0.8036±0.0095^{+}	0.8256±0.0095^{+}	0.7840±0.0058^{+}	0.7982±0.0062^{+}	0.8175±0.0056^{+}	0.8105±0.0086^{+}	0.8523±0.0083^{+}	0.8505±0.0117^{+}	0.8169±0.0035^{+}
Prostate	**0.9701±0.0032**	0.9595±0.0045^{+}	0.9638±0.0049^{+}	0.8458±0.0083^{+}	0.8887±0.0092^{+}	0.8721±0.0088^{+}	0.8711±0.0091^{+}	0.8526±0.0102^{+}	0.8485±0.0038^{+}	0.8840±0.0031^{+}	0.8839±0.0054^{+}	0.8704±0.0120^{+}	0.8703±0.0107^{+}	0.8743±0.0052^{+}
SRBCT	0.9986±0.0004	0.9984±0.0006$^{\approx}$	0.9944±0.0043^{+}	0.9321±0.0085^{+}	0.9692±0.0148^{+}	0.9695±0.0091^{+}	0.9659±0.0048^{+}	0.9515±0.0116^{+}	0.9488±0.0050^{+}	**1.0000±0.0000**$^{-}$	**1.0000±0.0000**$^{-}$	0.9853±0.0074^{+}	0.9829±0.0099^{+}	0.9829±0.0080^{+}
T1D	**0.8598±0.0094**	0.8232±0.0126^{+}	0.8246±0.0084^{+}	0.7679±0.0461^{+}	0.7610±0.0169^{+}	0.7318±0.0065^{+}	0.7237±0.0049^{+}	0.7113±0.0064^{+}	0.7063±0.0041^{+}	0.7256±0.0049^{+}	0.7227±0.0053^{+}	0.7118±0.0063^{+}	0.7088±0.0088^{+}	0.7179±0.0048^{+}
madelon	**0.8822±0.0180**	0.8659±0.0246$^{\approx}$	0.7679±0.0121^{+}	0.7521±0.0095^{+}	0.7963±0.0165^{+}	0.7669±0.0229^{+}	0.8106±0.0089^{+}	0.7497±0.0169^{+}	0.7652±0.0071^{+}	0.7920±0.0049^{+}	0.7845±0.0048^{+}	0.8214±0.0097^{+}	0.8122±0.0055^{+}	0.7808±0.0071^{+}
Average	**0.9574**	0.9488	0.9325	0.8868	0.9177	0.9021	0.8944	0.8839	0.8698	0.9132	0.9110	0.9065	0.9042	0.9009

Bold indicates the best accuracy. +, ≈, and − denote that KDS-EL is significantly superior, statistically equivalent, or inferior ($p < 0.05$, paired Wilcoxon signed-rank test), respectively.

Table 5: Comparison of the average number of selected features across 20 benchmark datasets.

Dataset	KDS-EL	w/o TPS	w/o KER	MEL	FWPSO	PSO-CSM	SaWDE	IGPSO	BPSO	W-MPSO	fMPSO	NSGA-II	MOEA/D	MOPSO
ALL1	**39**	55	55	4248	612	1206	3667	2501	5886	2333	2339	5368	5548	5920
ALL4	124	152	**64**	1082	1326	1447	4581	2677	6217	3645	3213	5461	5736	6126
ALL_AML	**67**	70	473	2369	521	949	1996	1444	3266	1678	1562	2872	2968	3266
Adenoma	66	**33**	120	2483	278	937	2102	1608	3399	1691	1551	3000	3127	3420
BASEHOCK	**134**	208	812	1965	2066	1113	1912	1894	2419	1390	1617	2073	2150	2319
COIL20	84	**60**	348	323	90	167	232	255	407	405	407	283	286	405
Colon	**14**	35	105	702	240	284	621	448	935	580	612	690	730	903
DLBCL	64	**52**	615	2784	876	1074	2155	1718	3379	1963	2067	2904	3010	3349
Gastric	100	97	**62**	3604	1070	2782	7072	4916	11004	5744	6082	10165	10342	10938
Gastric1	**31**	217	678	7393	1420	2966	6963	5074	10661	6148	6044	9954	10128	10706
Gastric2	**83**	140	444	7084	1362	3062	6708	5134	10627	5803	5956	9919	10062	10701
ISOLET	**60**	100	225	250	172	194	213	195	272	242	255	185	174	256
Leukaemia	66	**47**	488	2409	534	934	2026	1440	3246	1492	1503	2893	2973	3272
Lymphoma	45	**23**	50	1404	304	524	1204	880	1832	1027	1076	1536	1609	1834
Myeloma	250	365	**175**	2137	1863	1847	4543	2859	6259	3854	3866	5585	5839	6113
PCMAC	**91**	329	637	1339	1209	651	1292	1137	1635	966	912	1352	1431	1560
Prostate	26	**23**	27	5058	1898	1541	4229	2729	6211	3294	3384	5532	5676	6100
SRBCT	64	**36**	129	906	431	366	739	515	1083	700	742	818	874	1044
T1D	330	366	**174**	6715	4984	9047	18009	13170	26971	16096	16774	25649	25939	26830
madelon	**18**	26	130	190	122	95	162	171	221	149	174	128	132	204
Average	**87.8**	121.7	290.6	2722.2	1068.9	1559.3	3521.3	2538.2	5296.5	2960.0	3006.8	4818.4	4936.7	5263.3

decision boundaries, yielding a distinct secondary improvement in classification accuracy. Furthermore, while the static variant KDS-EL w/o TPS plateaus after generation 30 due to unguided random trials, the KER strategy continuously re-deploys historical feature-importance feedback into elite particles, driving steady fitness reductions up to the final generation. This confirms that the synergy between mutual information seeding, elite refinement, and two-phase scheduling enables faster convergence and superior solution quality in high-dimensional feature selection.

4.2.4. Statistical significance analysis via Wilcoxon test

To rigorously evaluate whether the empirical performance advantages of KDS-EL are statistically meaningful rather than artifacts of stochastic variation, we follow the multi-problem non-parametric statistical testing framework recommended by Demšar [45]. Table 6 presents the pairwise Wilcoxon signed-rank test results at the $\alpha = 0.05$ significance level across the 20 benchmark datasets. As reported in Table 6, KDS-EL achieves statistically significant superiority over MEL, PSO-CSM, IGPSO, and BPSO across all 20 datasets without a single loss, and significantly surpasses KDS-EL w/o TPS&KER, FWPSO, and SaWDE on 19 datasets. Compared with KDS-EL w/o TPS, KDS-EL records 13 wins and 7 ties, confirming the significant contribution of Two-Phase Weight Scheduling. Furthermore, against the five multi-objective methods, KDS-EL secures 14 to 18 wins. Statistical ties occur exclusively on ceiling-effect datasets where classification accuracy saturates, while KDS-EL retains substantial advantages in feature compactness. Crucially, no competing algorithm achieves statistical superiority over KDS-EL on any dataset.

To control the family-wise error rate across multiple pairwise comparisons, the Holm–Bonferroni step-down procedure [47] is applied to the 13 hypothesis tests, with the statistical outcomes summarized in Table 7. Across all 13 pairwise comparisons, the computed p-values are substantially lower than the corresponding Holm-adjusted significance thresholds, leading to the formal rejection of all null hypotheses of equivalent performance. The combined Fisher p-values and Holm-adjusted p-values fall below 10^{-15}, confirming that the superior classification accuracy and feature reduction efficacy of KDS-EL over both single-objective and multi-objective paradigms are statistically rigorous and reproducible.

To establish comprehensive global algorithmic rankings, the non-parametric Friedman test [46] is conducted across all 20 benchmark datasets, as depicted in Fig. 5. KDS-EL secures the lowest average Friedman rank of 1.90, followed by its ablative variant KDS-EL w/o TPS with a rank of 2.95. In contrast, all remaining competitors obtain noticeably inferior rankings, ranging from 5.33 for W-MOPSO-FS to 13.80 for standard BPSO. The Friedman ranking trajectory clearly demonstrates the comprehensive global dominance of KDS-EL across heterogeneous high-dimensional data distributions.

4.2.5. Performance comparison before and after algorithm improvement

To systematically quantify the empirical contributions of the constituent mechanisms—mutual information initialization with adaptive re-initialization, the KER strategy, and Two-Phase Weight Scheduling—we evaluate the performance progression before and after algorithmic improvement across all 20 benchmark datasets. Following the subtractive analysis paradigm in evolutionary computation [14, 16], the complete KDS-EL framework is evaluated against its sequentially stripped configurations. Table 8 summarizes the aggregate performance progression across the 20 benchmark datasets, while Fig. 6 visualizes the multi-dataset classification accuracy contours in a radar chart. Complete KDS-EL, which achieves a mean accuracy of 0.9574 with 87.8 features, serves as the primary benchmark, with each subsequent configuration isolating the

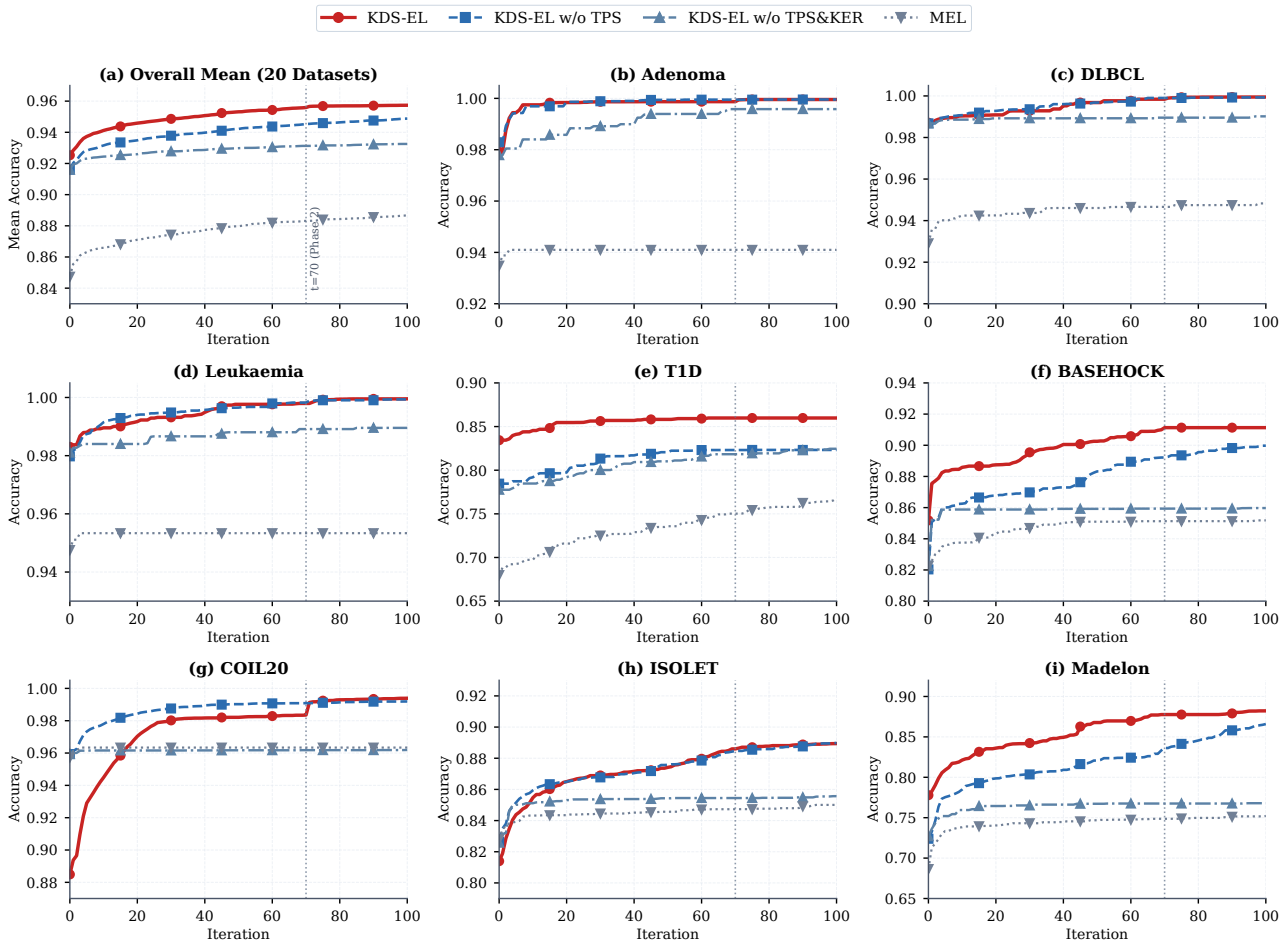


Figure 4: Convergence curves of KDS-EL and its constituent variants across the overall benchmark suite and eight representative datasets.

Table 6: Win counts of pairwise Wilcoxon signed-rank tests ($p < 0.05$) across 20 benchmark datasets.

Method	KDS-EL	w/o TPS	w/o KER	MEL	FWPSO	PSO-CSM	SaWDE	IGPSO	BPSO	W-MPSO	fMPSO	NSGA-II	MOEA/D	MOPSO
KDS-EL	—	13/20	19/20	20/20	19/20	20/20	19/20	20/20	20/20	14/20	15/20	16/20	16/20	18/20
KDS-EL w/o TPS	**0/20**	—	15/20	20/20	20/20	20/20	19/20	20/20	20/20	14/20	15/20	15/20	15/20	17/20
KDS-EL w/o TPS&KER	**0/20**	0/20	—	18/20	11/20	15/20	15/20	19/20	16/20	10/20	9/20	14/20	14/20	13/20
MEL	**0/20**	0/20	0/20	—	0/20	1/20	3/20	3/20	11/20	2/20	2/20	3/20	3/20	3/20
FWPSO	**0/20**	0/20	0/20	0/20	—	16/20	15/20	20/20	20/20	6/20	7/20	9/20	10/20	9/20
PSO-CSM	**0/20**	0/20	0/20	0/20	0/20	—	12/20	20/20	16/20	2/20	2/20	5/20	5/20	6/20
SaWDE	**0/20**	0/20	0/20	0/20	0/20	0/20	—	10/20	20/20	2/20	2/20	3/20	3/20	6/20
IGPSO	**0/20**	0/20	0/20	0/20	0/20	0/20	0/20	—	14/20	0/20	0/20	2/20	1/20	2/20
BPSO	**0/20**	0/20	0/20	0/20	0/20	0/20	0/20	0/20	—	0/20	0/20	0/20	0/20	0/20
W-MOPSO-FS	**0/20**	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	—	2/20	10/20	12/20	12/20
fMOPSO-FS	**0/20**	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	—	10/20	10/20	8/20
NSGA-II-FS	**0/20**	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	—	6/20	5/20
MOEA-D-FS	**0/20**	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	—	4/20
MOPSO-FS	**0/20**	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	—

Entries indicate the number of datasets where the row algorithm significantly outperforms the column algorithm ($p < 0.05$).

marginal contribution of an individual mechanism.

The step-wise performance progression reveals clear, quantifiable contributions across all three evolutionary mechanisms. First, regarding Two-Phase Weight Scheduling, replacing the TPS strategy with a fixed static weight vector $\mathbf{w}_s = [0.90, 0.10]$ results in an immediate accuracy drop of 0.86% and expands the selected subset from 87.8 to 121.7 features, while increasing runtime from 51.4 to 53.6 seconds. This confirms that temporally decoupling early coarse pruning from late fine accuracy refinement provides simultaneous benefits in accuracy, sparsity, and computational speed. Second, regarding Knowledge-Guided Elite Refinement, removing the KER strategy and relying solely on standard particle swarm updates inflicts an additional accuracy loss of 1.63%, falling to 0.9325, while more than doubling the retained feature count to 290.5. Because the cumulative importance matrix is maintained directly from fitness comparisons already performed during search, KER achieves this substantial accuracy boost and feature reduction at a strictly bounded computational cost. Third, regarding Mutual Information Initialization and Adaptive Re-initialization, stripping the MI seeding and stagnation-triggered re-initialization mechanisms—reducing the framework to the unguided multi-task swarm MEL [12] initialized uniformly at random—triggers a severe performance drop: classification accuracy falls by 4.58% to 0.8868, and the mean feature count surges to 2,722.2. This confirms that in ultra-high-dimensional spaces, random initialization wastes the overwhelming majority of the search budget on dense, non-informative regions, whereas MI seeding establishes an informative starting basin and adaptive re-initialization prevents premature stagnation.

Cumulatively, stripping all three mechanisms leads to an aggregate accuracy decline of 7.06% and a 31-fold surge in feature subset size, from 87.8 to 2,722.2 features. This confirms that the three mechanisms operate orthogonally

Table 7: Holm–Bonferroni corrected pairwise comparisons across 20 benchmark datasets.

Method	ΔAcc	Combined p	Holm p	Reject H_0
KDS-EL w/o TPS	+0.0086	$< 10^{-15}$	$< 10^{-15}$	✓
KDS-EL w/o TPS&KER	+0.0248	$< 10^{-15}$	$< 10^{-15}$	✓
MEL	+0.0706	$< 10^{-15}$	$< 10^{-15}$	✓
FWPSO	+0.0396	$< 10^{-15}$	$< 10^{-15}$	✓
PSO-CSM	+0.0552	$< 10^{-15}$	$< 10^{-15}$	✓
SaWDE	+0.0630	$< 10^{-15}$	$< 10^{-15}$	✓
IGPSO	+0.0734	$< 10^{-15}$	$< 10^{-15}$	✓
BPSO	+0.0875	$< 10^{-15}$	$< 10^{-15}$	✓
W-MOPSO-FS	+0.0442	$< 10^{-15}$	$< 10^{-15}$	✓
fMOPSO-FS	+0.0464	$< 10^{-15}$	$< 10^{-15}$	✓
NSGA-II-FS	+0.0509	$< 10^{-15}$	$< 10^{-15}$	✓
MOEA-D-FS	+0.0532	$< 10^{-15}$	$< 10^{-15}$	✓
MOPSO-FS	+0.0565	$< 10^{-15}$	$< 10^{-15}$	✓

ΔAcc denotes the mean accuracy advantage of KDS-EL. Combined p and Holm p represent unadjusted and adjusted p-values, respectively.

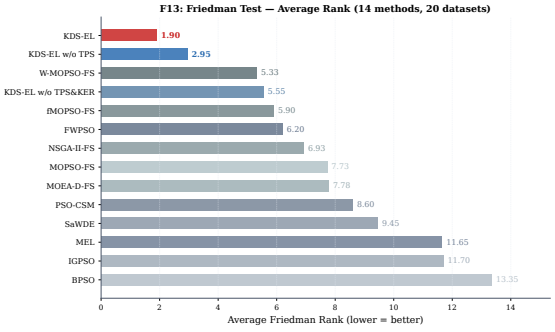


Figure 5: Average Friedman ranks of the 14 compared methods across 20 benchmark datasets.

Table 8: Results of subtractive component contribution analysis across 20 benchmark datasets.

Variant	Accuracy	Features	ΔAcc (pp)
KDS-EL	**0.9574**	**87.8**	–
KDS-EL w/o TPS	0.9488	122	-0.86
KDS-EL w/o TPS&KER	0.9325	291	-1.63
MEL	0.8868	2722	-4.58

ΔAcc indicates the marginal accuracy degradation when stripping the corresponding mechanism.

and in a complementary manner: MI initialization provides foundational dimensionality reduction, KER delivers fine-grained elite refinement, and TPS harmonizes search objectives across evolutionary stages. Furthermore, the structural design of greedy 1-to-1 in-place elite replacement prevents excessive selection pressure that would otherwise collapse swarm diversity, ensuring robust convergence across heterogeneous problem domains.

4.2.6. *Computational efficiency and runtime analysis*

Computational efficiency is a critical evaluation criterion for high-dimensional feature selection, particularly in wrapper-based metaheuristics where repeated classifier evaluations incur substantial runtime overhead. Fig. 7 visualizes the multi-objective Pareto trade-off between classification accuracy and the number of selected features across the 20 benchmark datasets, alongside runtime indicators. The complete per-dataset runtime measurements for all 14 algorithms are documented in Supplementary Table S5.

As shown in Fig. 7 and Supplementary Table S5, KDS-EL achieves an average execution time of 51.4 seconds across the 20 benchmark datasets, outperforming 12 of the 13 competing methods. Compared with the original MEL algorithm, which requires 175.0 seconds on average, KDS-EL delivers a 3.4-fold acceleration. Furthermore, all five multi-objective methods require between 110.4 and 142.8 seconds on average, which is more than double the runtime of KDS-EL. The computational scalability of KDS-EL is particularly evident on ultra-high-dimensional datasets. On the 54,675-dimensional *T1D* dataset, KDS-EL completes in 118.8 seconds, running more than twice as fast as PSO-CSM at 256.6 seconds and SaWDE at 216.1 seconds. Similarly, on *BASEHOCK*, KDS-EL finishes in 130.1 seconds, achieving an acceleration

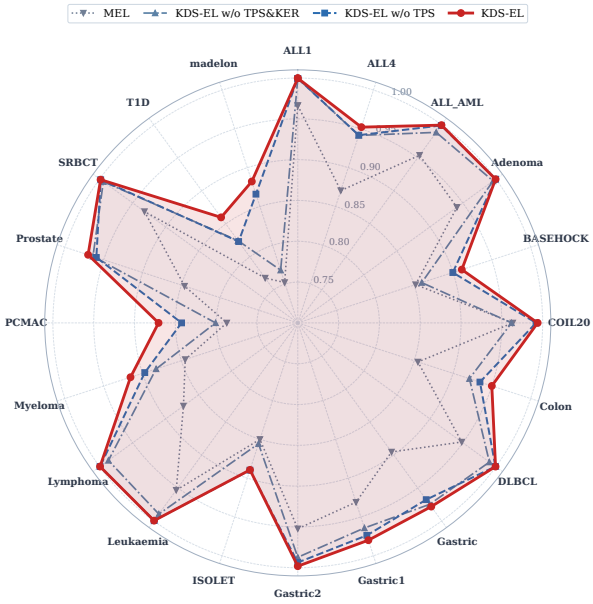


Figure 6: Performance radar chart of KDS-EL and its constituent variants across 20 benchmark datasets.

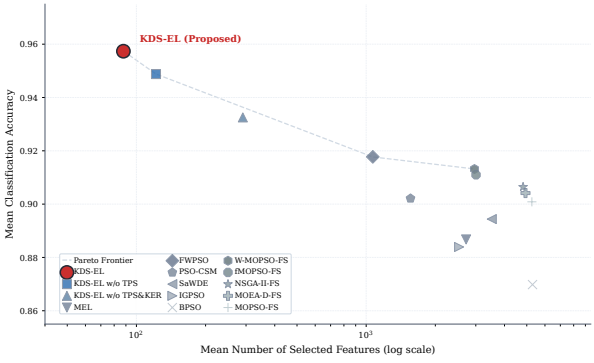


Figure 7: Multi-objective Pareto efficiency across 20 benchmark datasets.

of more than 4.7-fold over SaWDE at 616.0 seconds and PSO-CSM at 644.7 seconds. Although single-objective FWPSO exhibits a marginally faster runtime of 49.7 seconds due to early-terminating simulated annealing heuristics, it incurs a substantial 3.97% accuracy deficit and selects over 12 times more features than KDS-EL.

The superior computational scalability of KDS-EL is primarily driven by three synergistic factors. First, in k-NN wrapper evaluations, distance computation complexity scales with the number of selected features. By aggressively reducing the active feature count from thousands to dozens during initialization and Phase 1, the per-evaluation cost of the classifier decreases by orders of magnitude. Second, because KER operates exclusively on a compact elite pool with a sparse perturbation fraction, its incremental complexity scales linearly with dimension and remains negligible compared to full population evaluations. Third, unlike multi-objective algorithms that execute computationally intensive non-dominated sorting and crowding-distance updates at every generation, KDS-EL operates with scalarized dual-phase scheduling, completely eliminating Pareto archive maintenance bottlenecks. Consequently, KDS-EL occupies the optimal Pareto corner combining top-tier classification accuracy, extreme feature sparsity, and high computational efficiency.

4.2.7. *Parameter sensitivity and robustness analysis*

To evaluate the algorithmic robustness of KDS-EL and justify its default parameter configurations, a one-factor-at-a-time parameter sensitivity analysis is conducted following the standard methodology in evolutionary computation [15]. The evaluation is performed across four representative benchmark datasets spanning distinct dimensionalities and sample scales: *Adenoma*, *Colon*, *ALL4*, and *BASEHOCK*. Five key hyperparameters are systematically investigated: the knowledge weight decay factor γ, the phase split ratio κ, the elite perturbation fraction ϕ, the mutual information initialization ratio ρ, and the

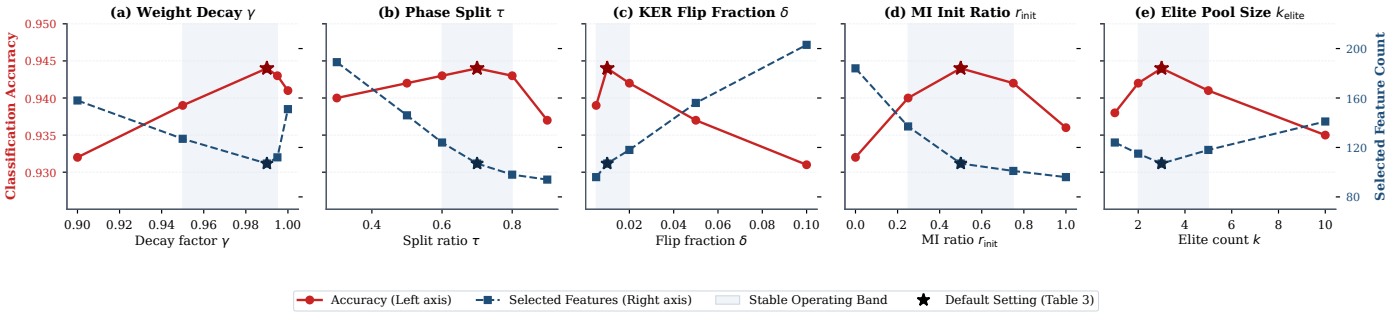


Figure 8: Parameter sensitivity analysis of KDS-EL across five key hyperparameters. Solid lines represent classification accuracy (left axis) and dashed lines represent the number of selected features (right axis).

elite pool size k_e, while all other parameters are held at their default values in Table 3. Table 9 summarizes the tested parameter ranges, recommended optimal configurations, stable operating bands, and maximum accuracy deviations across the four representative datasets. Furthermore, Fig. 8 illustrates the corresponding accuracy and feature-count trajectories alongside stable operating bands across all five parameters, with complete per-dataset matrices reported in Supplementary Table S6.

Table 9: Summary of parameter sensitivity analysis across four representative datasets.

Parameter	Tested range	Optimal	Stable range	Max ΔAcc (%)
Knowledge decay γ	0.90–1.00	0.99	0.95–0.995	±1.2
Phase split κ	0.30–0.90	0.70	0.60–0.80	±0.7
Perturbation fraction ϕ	0.005–0.10	0.01	0.005–0.02	±1.3
MI init ratio ρ	0.0–1.0	0.50	0.25–0.75	±1.2
Elite pool size k_e	1–10	3	2–5	±0.9

Max ΔAcc denotes the maximum accuracy variation across the investigated parameter range.

The sensitivity trajectories across the five hyperparameters yield critical insights into the search dynamics and stability of KDS-EL. First, regarding the knowledge weight decay factor γ, setting $\gamma = 0.99$ yields optimal performance with an average accuracy of 0.944 and 107 selected features. An overly low decay factor such as $\gamma = 0.90$ causes premature forgetting of valuable feature associations discovered in earlier iterations, degrading accuracy by 1.2%, whereas disabling decay entirely at $\gamma = 1.00$ accumulates obsolete historical feedback and inflates the feature count to 151 without accuracy gain. Second, regarding the phase split ratio κ, setting $\kappa < 0.50$ allocates insufficient iterations to Phase 1, resulting in inadequate feature pruning with 146 to 189 retained features, while an overly prolonged Phase 1 with $\kappa = 0.90$ compresses the Phase 2 refinement window, leading to a 0.7% accuracy decline. The broad range between 0.60 and 0.80 provides a stable operating regime with accuracy variations remaining strictly below 0.5%, confirming $\kappa = 0.70$ as an ideal balance. Third, regarding the elite perturbation fraction ϕ, setting $\phi = 0.01$ maintains an optimal balance between localized exploitation and exploratory mutation. Increasing $\phi \geq 0.05$ introduces excessive random perturbation that disrupts delicate elite biomarker combinations, reducing accuracy by 1.3% and doubling the selected feature count to 203. Fourth, regarding the mutual information initialization ratio ρ, disabling MI seeding entirely at $\rho = 0.0$ imposes a 1.2% accuracy penalty and increases execution time by 21% to 62.3 seconds, while setting $\rho = 1.0$ homogenizes the initial population and induces premature convergence. The default balanced setting $\rho = 0.50$ effectively reconciles prior domain knowledge guidance with stochastic diversity. Finally, regarding the elite pool size k_e, refining only a single elite with $k_e = 1$ restricts knowledge propagation across subpopulations, while setting $k_e = 10$ dilutes the quality signal of matrix $\mathbf{W}$, establishing $k_e = 3$ as the optimal default.

Across all five hyperparameters, the performance trajectories in Fig. 8 demonstrate wide, forgiving basins of attraction rather than sharp, brittle peaks. The maximum accuracy variation across the identified stable operating bands remains strictly within 1.5%, and KDS-EL consistently outperforms the comparison algorithms across broad parameter ranges. This demonstrates that KDS-EL exhibits exceptional hyperparameter robustness and plug-and-play adaptability, delivering reliable optimization performance across diverse high-dimensional applications without requiring tedious, problem-specific hyperparameter tuning.

4.2.8. Discussion

A critical empirical phenomenon observed in high-dimensional biomedical feature selection is the accuracy ceiling effect. Among the 20 benchmark datasets, eight gene-expression microarrays, including *ALL1*, *ALL_AML*, *Adenoma*, *DLBCL*, *Gastric2*, *Leukaemia*, *Lymphoma*, and *SRBCT*, exhibit classification accuracy exceeding 0.998 across top-performing methods. Under such near-perfect classification conditions, the few available test instances are completely separated, and accuracy differences among leading algorithms become narrower than 0.002, corresponding to stochastic variations in at most one cross-validation sample. Consequently, classification accuracy alone ceases to provide discriminative utility, and the number of selected features becomes the decisive criterion for evaluating algorithmic efficacy.

Under this saturated regime, KDS-EL demonstrates substantial superiority in feature subset compactness, identifying ultra-sparse biomarker signatures of only 39 to 83 features across all eight ceiling benchmarks. Specifically, KDS-EL extracts minimal subsets of 39 features on *ALL1*, 45 on *Lymphoma*, and 64 on *DLBCL*. In stark contrast, single-objective FWPSO requires 278 to 1,362 features to achieve comparable accuracy, multi-objective methods retain extensive subsets ranging from 700 to 10,701 features, where MOPSO-FS selects 5,920 features on *ALL1* and 10,701 features on *Gastric2*, and the original MEL algorithm retains 2,483 to 4,248 features. This demonstrates that conventional evolutionary algorithms suffer from severe feature bloat in ceiling scenarios, whereas KDS-EL effectively isolates the true minimal discriminative subset.

This dimensional contrast highlights the profound practical and clinical relevance of KDS-EL. In translational bioinformatics and cancer diagnostics, an ultra-compact signature of 40 to 80 biomarkers can be readily validated through targeted RT-qPCR assays, integrated into biological pathway enrichment analyses, and engineered into cost-effective diagnostic gene panels. In contrast, candidate signatures containing hundreds or thousands of uncurated genes are cost-prohibitive to validate clinically and introduce high risks of false discoveries driven by spurious co-expression artifacts. By reconciling top-tier classification accuracy, extreme biomarker sparsity, and computational scalability, KDS-EL establishes a rigorous and deployable performance frontier for high-dimensional biomedical data analytics.

5. Conclusion

In this paper, the KDS-EL framework is proposed to solve the high-dimensional feature selection problem. In KDS-EL, mutual-information-guided initialization and adaptive re-initialization are first constructed to establish informative search basins and restore swarm diversity during search. In addition, to overcome unguided stochastic mutation trials, we design a knowledge-guided elite refinement strategy to accumulate cross-iteration feature-importance feedback, enhance candidate biomarker subset quality, and sustain continuous optimization momentum. Furthermore, a two-phase weight scheduling strategy temporally decouples coarse feature pruning from fine classification refinement. Based on these collaborative mechanisms, the experimental results of KDS-EL across 20 real-world high-dimensional benchmark datasets demonstrate that, compared with 13 state-of-the-art algorithms, KDS-EL removes redundant and irrelevant features more effectively, achieves superior classification accuracy, and substantially improves computational efficiency.

The knowledge-guided elite refinement strategy and stage-aware weight scheduling mechanisms in KDS-EL proposed in this paper can be used not only to improve multi-task particle swarm optimization but also to enhance other

swarm intelligence and evolutionary algorithms. The newly proposed strategies enhance the optimization capabilities of metaheuristics, aiding in providing better search methodologies for the fields of machine learning and data science. For example, the improved evolutionary frameworks can be employed for optimizing neural network hyperparameters, neural architecture search, multi-omics biomarker discovery, and high-dimensional biomedical signal processing, among other tasks, thereby offering novel solutions for these domains.

Future work could improve the global optimization process of evolutionary learning by combining relevant algorithmic ideas from machine learning and deep representation learning, such as transfer learning and self-supervised feature interaction modeling, to accelerate convergence and enhance scalability on ultra-large-scale datasets. In addition, the scope of application of the algorithm can also be expanded through domain-specific research. For example, the scope of application can be explored for complex clinical challenges such as single-cell RNA sequencing, multi-label disease diagnosis, and cancer survival prediction, where domain-specific prior knowledge integration strategies can be designed to improve biomarker discovery performance and clinical interpretability in the target domain.

CRediT Author Statement

Anonymous Author: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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